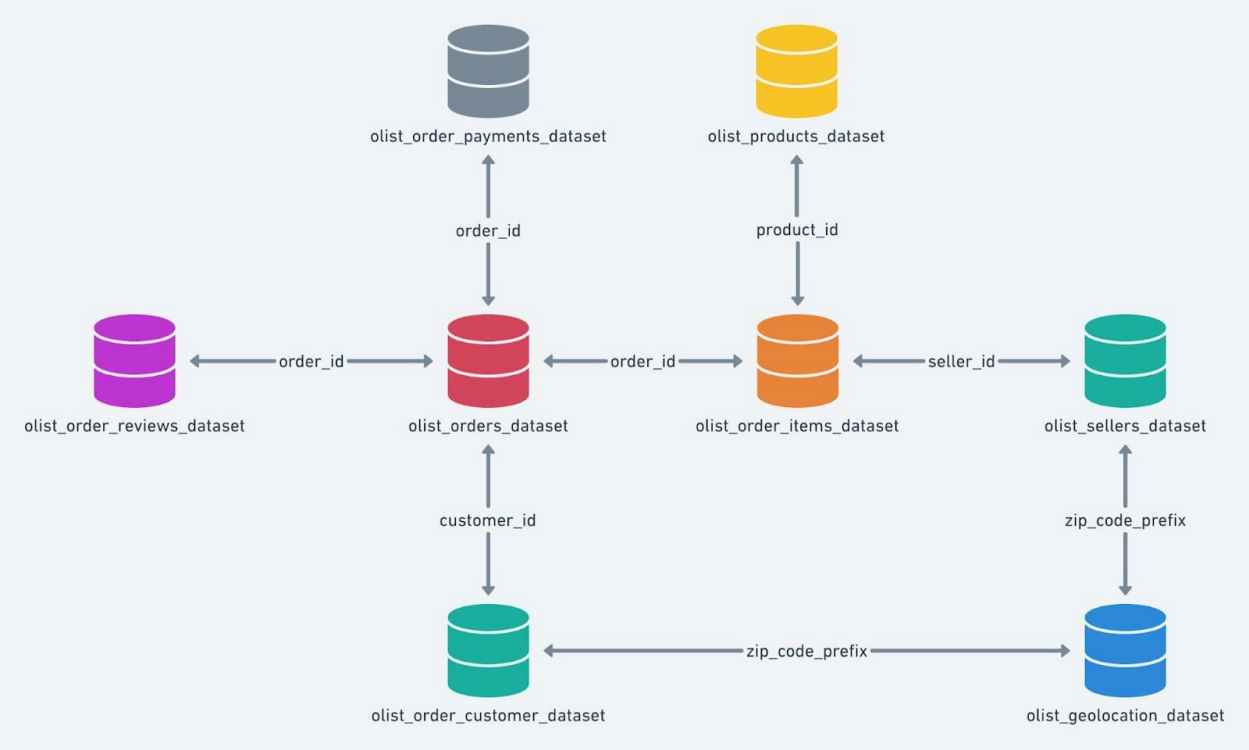
E-commerce Data Analysis(Target Dataset)



- STEP 1: **EDA**

Understanding the dataset is crucial. So, let's begin with the table customers.

#Customers:

Customers table has columns:

1. customer_id
2. customer_unique_id
3. customer_zipcode
4. customer_city
5. customer_state

Customer_id is associated with the orders table and customer_unique_id is Unique identification of the customer given to each customer. So, this is a clear example of the Primary key and UNIQUE key.

- Understanding the Data Types of all columns present in customers' table.

```
SELECT
  column_name,
  data_type
FROM `project1-487022.Project1.INFORMATION_SCHEMA.COLUMNS`
WHERE table_name = 'customers';
```

---

column_name	data_type
customer_id	STRING
customer_unique_id	STRING
customer_zip_code	INT64
customer_city	STRING
customer_state	STRING

---

2. Let's try to understand “Time between the orders were Placed”.

```
SELECT
  MIN(TIME(order_purchase_timestamp)) as StartTime_OrderPurchase,
  Max(TIME(order_purchase_timestamp)) as EndTime_orderPurchase
FROM `Project1.orders`;
```

Row	StartTime_OrderP...	EndTime_orderP...
1	00:00:00	23:59:59

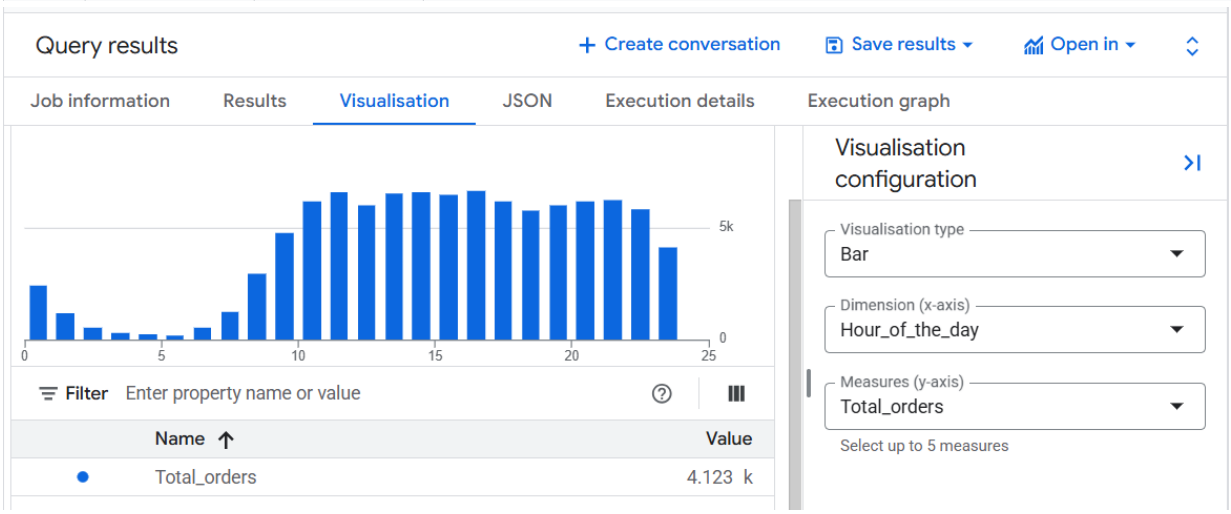
because this is a massive e-commerce dataset, **at least one order** has been placed at essentially every second of the day.

Seeing `00:00:00` to `23:59:59` tells you the "boundary," but it doesn't tell you the **story** of when your customers are actually awake and buying.

To find the **actual time window** (the peak hours), we need to aggregate the data.

```
SELECT
  EXTRACT(HOUR FROM order_purchase_timestamp) as Hour_of_the_day,
  COUNT(*) as Total_orders
FROM `Project1.orders`
GROUP BY 1
ORDER BY 2 DESC;
```

Row	Hour_of_the_day	Total_orders
1	16	6675
2	11	6578
3	14	6569
4	13	6518
5	15	6454



Key Insights from Bar Chart

- **The "Active" Window:** While orders happen 24/7, the significant volume starts climbing around **08:00 (8 AM)** and stays high until **22:00 (10 PM)**.
- **The "Lunchtime Peak":** Your table shows the top hour is **16:00 (4 PM)**, followed closely by **11:00 (11 AM)**. This suggests people are shopping during their work breaks or right before heading home.
- **The "Late Night Slump":** There is a sharp drop-off after **midnight until 6 AM**, which is typical for most retail markets.

Turning Insights into Business Action

1. **Customer Support:** Ensure maximum staff are online between **10:00 and 17:00**.
2. **Ad Spend:** Increase your marketing budget during the **11:00 – 16:00** window.

Now that we know **when** people shop, Let's see **where** they shop from during those peak hours?

Top 3 States for orders placed during the 16:00 (Peak) hour.

```
SELECT
  c.customer_state,
  COUNT(*) as Total_orders
FROM `Project1.orders` as o
LEFT JOIN `Project1.customers` as c
ON o.customer_id = c.customer_id
WHERE EXTRACT(HOUR FROM o.order_purchase_timestamp) = 16
GROUP BY c.customer_state
ORDER BY 2 DESC
LIMIT 3;
```

Row	customer_state	Total_orders
1	SP	2811
2	RJ	858
3	MG	778

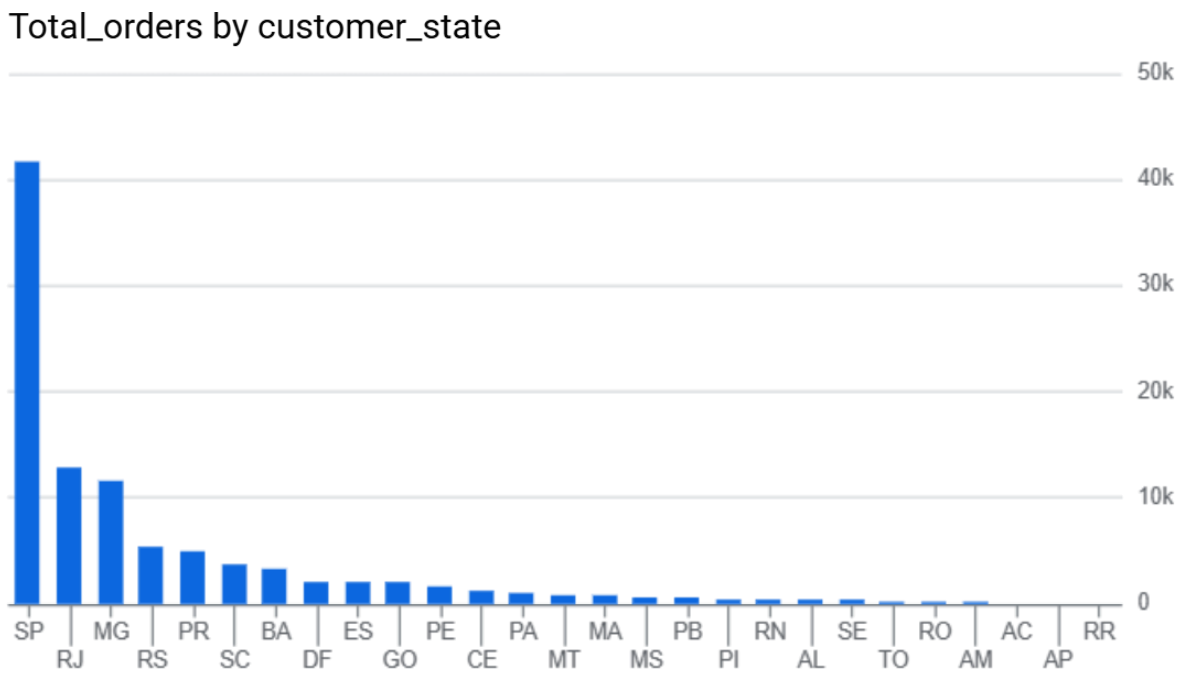
3 States where the number of orders is at least during the 16:00(Peak) Hour.

Row	customer_state	Total_orders
1	RR	5
2	AC	7
3	AP	9

"Percentage of Total Orders" each state contributes?

```
SELECT
  c.customer_state,
  COUNT(*) as Total_orders,
  ROUND(SAFE_DIVIDE(COUNT(*), SUM(COUNT(*)) OVER()) * 100,2) as
percentage_contribution
FROM `Project1.orders` as o
LEFT JOIN `Project1.customers` as c
ON o.customer_id = c.customer_id
GROUP BY c.customer_state
ORDER BY 3 DESC;
```

Row	customer_state	Total_orders	percentage_contr...
1	SP	41746	41.98
2	RJ	12852	12.92
3	MG	11635	11.7
4	RS	5466	5.5
5	PR	5045	5.07
6	SC	3637	3.66
7	BA	3380	3.4
8	DF	2140	2.15
9	ES	2033	2.04
10	GO	2020	2.03
11	PE	1652	1.66
12	CE	1336	1.34
13	PA	975	0.98
14	MT	907	0.91
15	MA	747	0.75
16	MS	715	0.72
17	PB	536	0.54
18	PI	495	0.5
19	RN	485	0.49
20	AL	413	0.42
21	SE	350	0.35
22	TO	280	0.28
23	RO	253	0.25
24	AM	148	0.15
25	AC	81	0.08
26	AP	68	0.07
27	RR	46	0.05



Executive Insights: Customer Order Behavior Analysis

Geographic Market Concentration

The data reveals a highly concentrated market, with a significant "long tail" distribution across Brazilian states.

- **Core Market Dominance: São Paulo (SP)** is the primary driver of business, contributing over **40% of total order volume**. This indicates a mature market presence and high brand penetration in this region.
- **The "Big Three" Hubs:** Together, **SP, RJ (Rio de Janeiro), and MG (Minas Gerais)** account for the vast majority of revenue. Operations and marketing efforts in these three states are critical to overall business stability.
- **Strategic Opportunity:** The sharp decline in order volume across the remaining 20+ states suggests significant room for growth. States like **AC (Acre) and RR (Roraima)** show minimal activity, which may point to high shipping costs, long lead times, or low brand awareness in Northern regions.

Now that we know **who** is buying (States) and **when** they are buying (Time), we might want to look at **what** they are buying or **how much** they are spending.

Q. Product Category Analysis: (Helpful for marketing and warehouse strategy)

```
SELECT
  `product category` as product_category,
  COUNT(*) as Total_order_quantity
FROM `Project1.orderItems` as oi
LEFT JOIN `project1-487022.Project1.products` as p
  ON oi.product_id = p.product_id
GROUP BY `product category`
ORDER BY 2 DESC;
```

product_category	Total_order_quantity
bed table bath	11115
HEALTH BEAUTY	9670
sport leisure	8641
Furniture Decoration	8334
computer accessories	7827
housewares	6964
Watches present	5991
telephony	4545
Garden tools	4347
automotive	4235
toys	4117
Cool Stuff	3796
perfumery	3419
babies	3065
electronics	2767
stationary store	2517
Fashion Bags and Accessories	2031

pet Shop 1947
Furniture office 1691
1603
Games consoles 1137
Bags Accessories 1092
Construction Tools Construction 929
home appliances 771
musical instruments 680
electrostile 679
Casa Construcao 604
General Interest Books 553
foods 510
Room Furniture 503
House comfort 434
drinks 379
audio 364
Market Place 311
Construction Tools Illumination 304
climatization 297
Furniture Kitchen Service Area Dinner and Garden 281
Drink foods 278
Industry Commerce and Business 268
technical books 267
fixed telephony 264
Fashion Calcados 262
Construction Tools Garden 238
ELECTRICES 2 238

Agro Industria e Comercio	212
Art	209
PCs	203
SIGNALIZATION AND SAFETY	199
CONSTRUCTION SECURITY TOOLS	194
Christmas articles	153
Fashion Men's Clothing	132
Fashion Underwear and Beach Fashion	131
Furniture	109
Construction Tools Tools	103
IMAGE IMPORT TABLETS	83
HOUSE PASTALS OVEN AND CAFE	76
cine photo	72
Blu Ray DVDs	64
Imported books	60
Fashion Women's Clothing	48
party articles	43
Hygiene diapers	39
CITTE AND UPHACK FURNITURE	38
song	38
flowers	33
House Comfort 2	30
Fashion Sport	30
Arts and Crafts	24
Kitchen portable and food coach	15
La Cuisine	14
cds music dvds	14

PC Gamer	9
Fashion Children's Clothing	8
insurance and services	2

Analysis C Insights :

Segment	Categories	Characterization
Powerhouses	Top 5 (Bed/Bath to Comp Accessories)	**High Volume:** Drives the majority of shipping traffic.
Mid-Tier	Watches, Toys, Automotive	**Specialty Interests:** Higher profit margins, lower frequency.
Niche/Long Tail	PC Gamer, Insurance, Children's Clothing	**Low Demand:** Might not be worth the warehouse shelf space.

Interesting Anomalies

- **Watches s Presents (#7):** This is surprisingly high! Usually, jewelry is a niche, but here it beats out "Toys" and "Electronics." This suggests your customers might be using the platform heavily for **Gifting**.
- **Insurance and Services (#Last):** With only 2 orders, this product category is a "Failure." Target would likely de-list this service to save on administrative costs.

Our business is heavily driven by **Home s Self-Care**. The top 10 categories account for a disproportionate amount of our logistics volume. However, we have a data integrity gap with over 1,600 unclassified orders that could be hiding further trends.

In-depth Exploration

1. Is there a growing trend in the no. of orders placed over the past years?

```
SELECT
  EXTRACT(YEAR FROM order_purchase_timestamp) as YEAR,
```

```
  COUNT(order_id) as Total_orders
FROM `Project1.orders`
GROUP BY EXTRACT(YEAR FROM order_purchase_timestamp)
ORDER BY YEAR ASC;
```

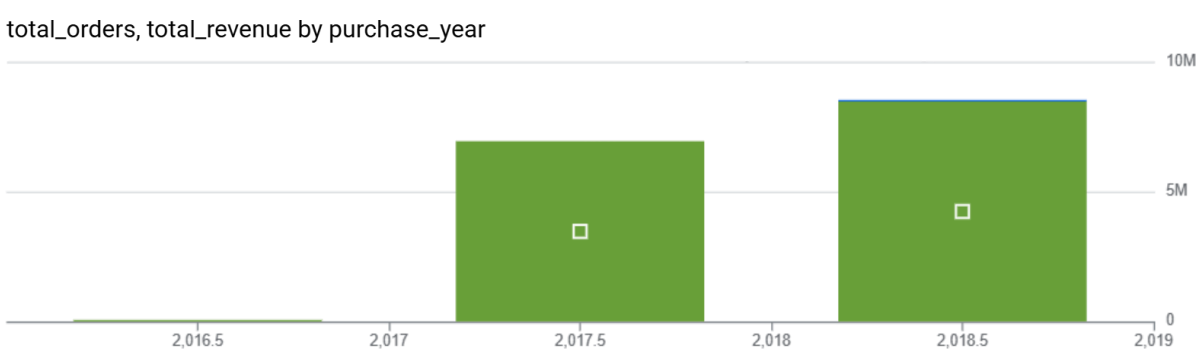
Row	YEAR	Total_orders
1	2016	329
2	2017	45101
3	2018	54011

Note: These are just the orders being placed

We can find how many of them being **delivered successfully** C **Total_Revenue**

```
SELECT
    EXTRACT(YEAR FROM o.order_purchase_timestamp) AS purchase_year,
    COUNT(DISTINCT o.order_id) AS total_orders,
    ROUND(SUM(p.payment_value), 2) AS total_revenue
FROM `Project1.orders` AS o
JOIN `Project1.payments` AS p
    ON o.order_id = p.order_id
WHERE o.order_status = 'delivered' -- We only count successful sales
GROUP BY 1
ORDER BY 1;
```

Purchase Year	Total Orders	Total Revenue	Average Order Value (AOV)
2016	266	$46,586.33	**$175.14**
2017	43,428	$6,922,900.24	**$15G.41**
2018	52,783	$8,452,975.20	**$160.15**



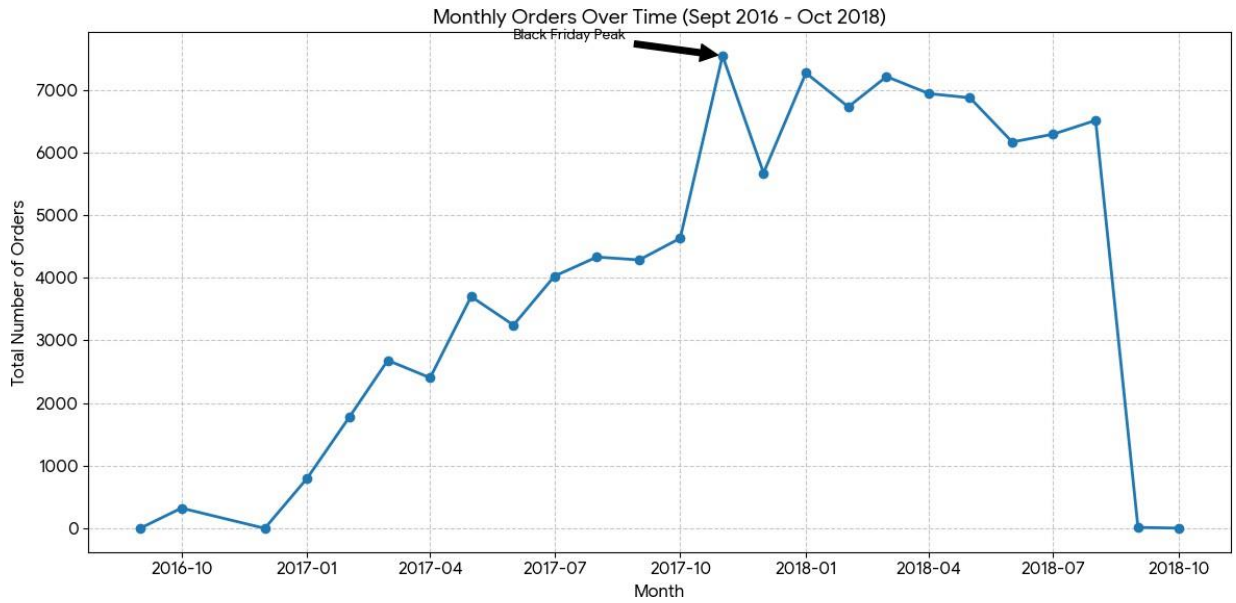
2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

```sql
select
  EXTRACT(YEAR from order_purchase_timestamp) as year,
  EXTRACT(MONTH from order_purchase_timestamp) as month,
  COUNT(distinct order_id)
FROM `Project1.orders`
GROUP BY year, month
ORDER BY year, month;
```

Row	year	month	Total_No_Orders
1	2016	9	4
2	2016	10	324
3	2016	12	1
4	2017	1	800
5	2017	2	1780
6	2017	3	2682
7	2017	4	2404
8	2017	5	3700
9	2017	6	3245
10	2017	7	4026

Row	year	month	Total_No_Orders
11	2017	8	4331
12	2017	9	4285
13	2017	10	4631
14	2017	11	7544
15	2017	12	5673
16	2018	1	7269
17	2018	2	6728
18	2018	3	7211
19	2018	4	6939
20	2018	5	6873

Row	year	month	Total_No_Orders
20	2018	5	6873
21	2018	6	6167
22	2018	7	6292
23	2018	8	6512
24	2018	9	16
25	2018	10	4



Key Insights: Orderlist Monthly Order Trends

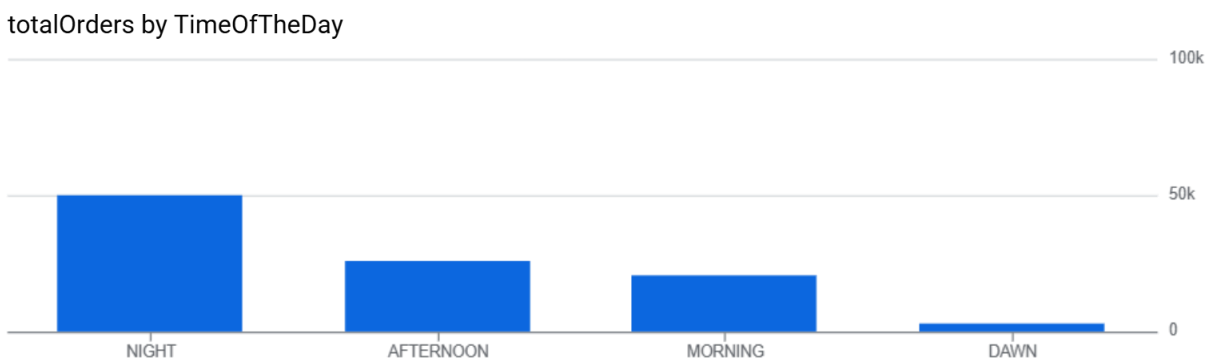
- **Explosive Growth:** The platform transitioned from a pilot phase in 2016 to a high-volume scale in 2017, with monthly orders peaking at **7,544** in November.
- **Black Friday Dominance:** November 2017 shows a massive **63% month-over-month spike**, identifying it as the critical annual sales driver for the business.
- **Market Maturity:** By 2018, the business stabilized at a "new normal" of **6,000–7,000 orders per month**, indicating a shift from rapid acquisition to steady-state operations.
- **Seasonal Dips:** A consistent softening occurs in **April and May** each year, suggesting a mid-year lull where promotional intervention might be needed.
- **Data Integrity:** Significant drops in September 2016 and September 2018 are **data collection artifacts** (incomplete months) rather than actual crashes in consumer demand.

3. what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)
    a. 0-6 hrs : Dawn
    b. 7-12 hrs : Mornings
    c. 13-18 hrs : Afternoon

d. 19-23 hrs : Night

```
SELECT
  t1.TimeOfTheDay,
  COUNT(DISTINCT t1.order_id) as totalOrders
FROM
(select
  *,
  CASE
  WHEN 0 < EXTRACT(HOUR from order_purchase_timestamp) AND  EXTRACT(HOUR from
order_purchase_timestamp) < 7 THEN "DAWN"
  WHEN 7 < EXTRACT(HOUR from order_purchase_timestamp) AND  EXTRACT(HOUR from
order_purchase_timestamp) < 12 THEN "MORNING"
  WHEN 13 < EXTRACT(HOUR from order_purchase_timestamp) AND  EXTRACT(HOUR from
order_purchase_timestamp) < 18 THEN "AFTERNOON"
  ELSE "NIGHT"
  END AS TimeOfTheDay
FROM `Project1.orders`) as t1
GROUP BY t1.TimeOfTheDay
ORDER BY 2 DESC;
```

Row	TimeOfTheDay	totalOrders
1	NIGHT	50238
2	AFTERNOON	25848
3	MORNING	20507
4	DAWN	2848



1. Chronological Order Trends (2016–2018)

The dataset reveals a significant transition from an initial market entry phase to a high-growth scaling period.

- **Expansion Dynamics:** Monthly order volume surged from approximately 800 units in January 2017 to over 7,200 units by January 2018, representing an 800% year-over-year increase.
- **Peak Performance:** November 2017 stands as the historical maximum with **7,544 orders**, driven primarily by the high-impact "Black Friday" promotional window.

- **Market Stabilization:** Throughout 2018, the platform maintained a consistent operational baseline, averaging between **6,000 and 7,000 orders monthly** until the data cutoff in September.

2. Diurnal Purchasing Behavior

Analysis of timestamps indicates distinct consumer habits based on the time of day.

- **Dominant Sales Window:** The **NIGHT** period is the primary driver of engagement, accounting for **50,238 total orders**.
- **Secondary Peak:** The **AFTERNOON** period follows with **25,848 orders**, suggesting that over 75% of total platform activity occurs after 12:00 PM.
- **Low-Activity Periods: DAWN** remains the lowest performing segment with only **2,848 orders**, indicating minimal late-night or early-morning conversion.

3. Strategic Observations

- **Seasonal Volatility:** Despite the overall growth, the platform experiences consistent "softening" in **April and May**, identifying a need for mid-year promotional interventions.
- **Data Integrity Note:** Sharp declines in September 2016 and September 2018 are attributed to **incomplete data collection cycles** at the edges of the dataset rather than a collapse in consumer demand.

Evolution of E-commerce orders in the Brazil region

1. Get the month on month no. of orders placed in each state.

```
select
  c.customer_state as state,
  EXTRACT(MONTH from o.order_purchase_timestamp) as months,
  COUNT(*) as totalOrders
from `Project1.orders` as o
left join `Project1.customers` as c
on o.customer_id = c.customer_id
GROUP BY months, state
ORDER BY state, months;
```

state	month	totalOrders
AC	1	8
AC	2	6
AC	3	4
AC	4	9
AC	5	10
AC	6	7
AC	7	9
AC	8	7
AC	9	5
AC	10	6
AC	11	5
AC	12	5
AL	1	39
AL	2	39
AL	3	40
AL	4	51
AL	5	46
AL	6	34
AL	7	40
AL	8	34
AL	9	20
AL	10	30
AL	11	26
AL	12	14

AM	1	12
AM	2	16
AM	3	14
AM	4	19
AM	5	19
AM	6	8
AM	7	23
AM	8	9
AM	9	9
AM	10	3
AM	11	10
AM	12	6
AP	1	11
AP	2	4
AP	3	8
AP	4	5
AP	5	11
AP	6	4
AP	7	7
AP	8	5
AP	9	2
AP	10	3
AP	11	4
AP	12	4
BA	1	264
BA	2	273

BA	3	340
BA	4	318
BA	5	368
BA	6	307
BA	7	405
BA	8	323
BA	9	170
BA	10	170
BA	11	250
BA	12	192
CE	1	99
CE	2	101
CE	3	126
CE	4	143
CE	5	136
CE	6	121
CE	7	140
CE	8	130
CE	9	77
CE	10	74
CE	11	108
CE	12	81
DF	1	151
DF	2	196
DF	3	207
DF	4	183

DF	5	208
DF	6	220
DF	7	243
DF	8	232
DF	9	97
DF	10	104
DF	11	168
DF	12	131
ES	1	159
ES	2	186
ES	3	182
ES	4	188
ES	5	228
ES	6	204
ES	7	206
ES	8	200
ES	9	93
ES	10	104
ES	11	170
ES	12	113
GO	1	164
GO	2	176
GO	3	199
GO	4	177
GO	5	226
GO	6	184

GO	7	192
GO	8	213
GO	9	88
GO	10	117
GO	11	157
GO	12	127
MA	1	66
MA	2	67
MA	3	77
MA	4	73
MA	5	65
MA	6	59
MA	7	79
MA	8	70
MA	9	42
MA	10	52
MA	11	56
MA	12	41
MG	1	971
MG	2	1063
MG	3	1237
MG	4	1061
MG	5	1190
MG	6	1080
MG	7	1111
MG	8	1177

MG	9	511
MG	10	600
MG	11	943
MG	12	691
MS	1	71
MS	2	75
MS	3	79
MS	4	58
MS	5	74
MS	6	76
MS	7	74
MS	8	59
MS	9	33
MS	10	34
MS	11	46
MS	12	36
MT	1	96
MT	2	84
MT	3	71
MT	4	92
MT	5	104
MT	6	83
MT	7	85
MT	8	78
MT	9	35
MT	10	55

MT	11	74
MT	12	50
PA	1	82
PA	2	83
PA	3	109
PA	4	107
PA	5	75
PA	6	92
PA	7	96
PA	8	104
PA	9	41
PA	10	58
PA	11	70
PA	12	58
PB	1	33
PB	2	47
PB	3	55
PB	4	51
PB	5	47
PB	6	51
PB	7	79
PB	8	46
PB	9	29
PB	10	31
PB	11	30
PB	12	37

PE	1	113
PE	2	146
PE	3	153
PE	4	154
PE	5	174
PE	6	140
PE	7	210
PE	8	170
PE	9	76
PE	10	87
PE	11	126
PE	12	103
PI	1	55
PI	2	46
PI	3	48
PI	4	50
PI	5	56
PI	6	43
PI	7	52
PI	8	43
PI	9	23
PI	10	25
PI	11	31
PI	12	23
PR	1	443
PR	2	460

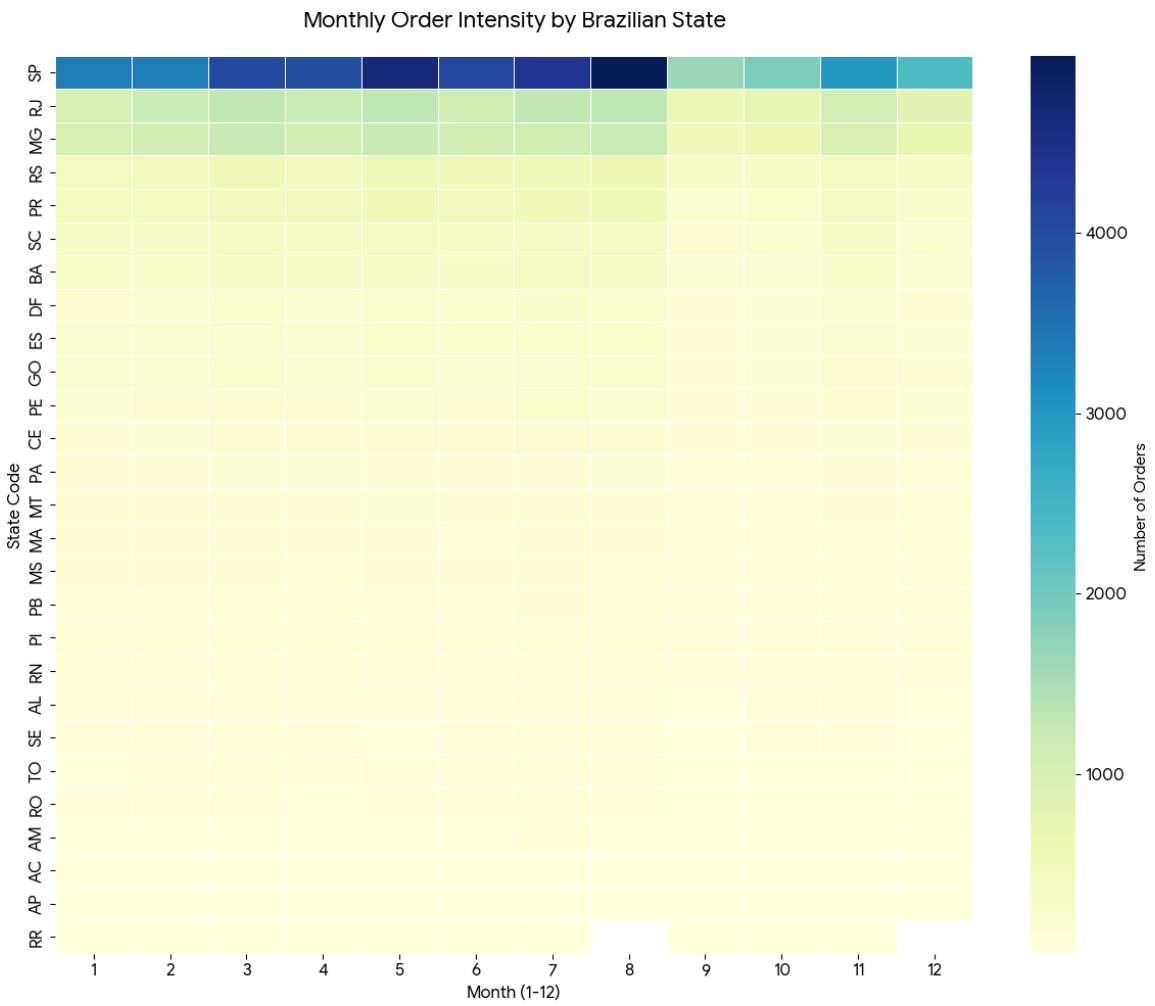
PR	3	504
PR	4	500
PR	5	524
PR	6	478
PR	7	523
PR	8	556
PR	9	183
PR	10	225
PR	11	378
PR	12	271
RJ	1	990
RJ	2	1176
RJ	3	1302
RJ	4	1172
RJ	5	1321
RJ	6	1128
RJ	7	1288
RJ	8	1307
RJ	9	612
RJ	10	725
RJ	11	1048
RJ	12	783
RN	1	51
RN	2	31
RN	3	52
RN	4	42

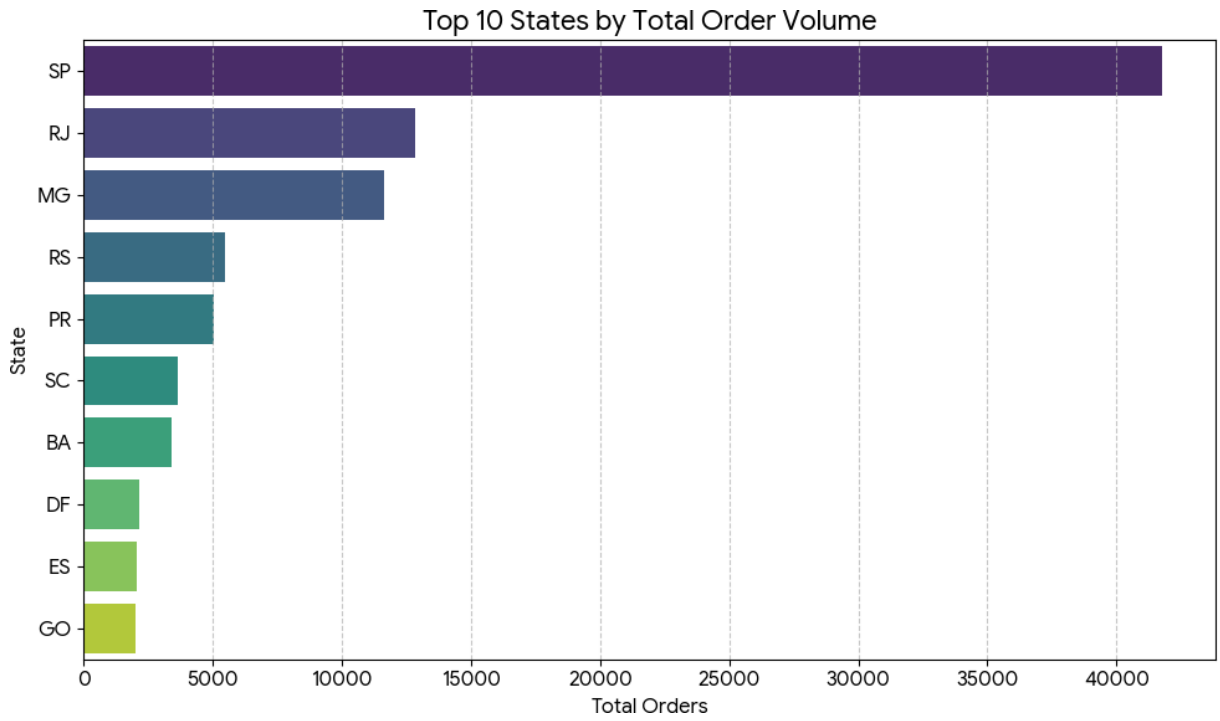
RN	5	39
RN	6	49
RN	7	56
RN	8	40
RN	9	24
RN	10	27
RN	11	44
RN	12	30
RO	1	23
RO	2	25
RO	3	29
RO	4	20
RO	5	26
RO	6	22
RO	7	27
RO	8	23
RO	9	16
RO	10	14
RO	11	17
RO	12	11
RR	1	2
RR	2	7
RR	3	8
RR	4	4
RR	5	3
RR	6	8

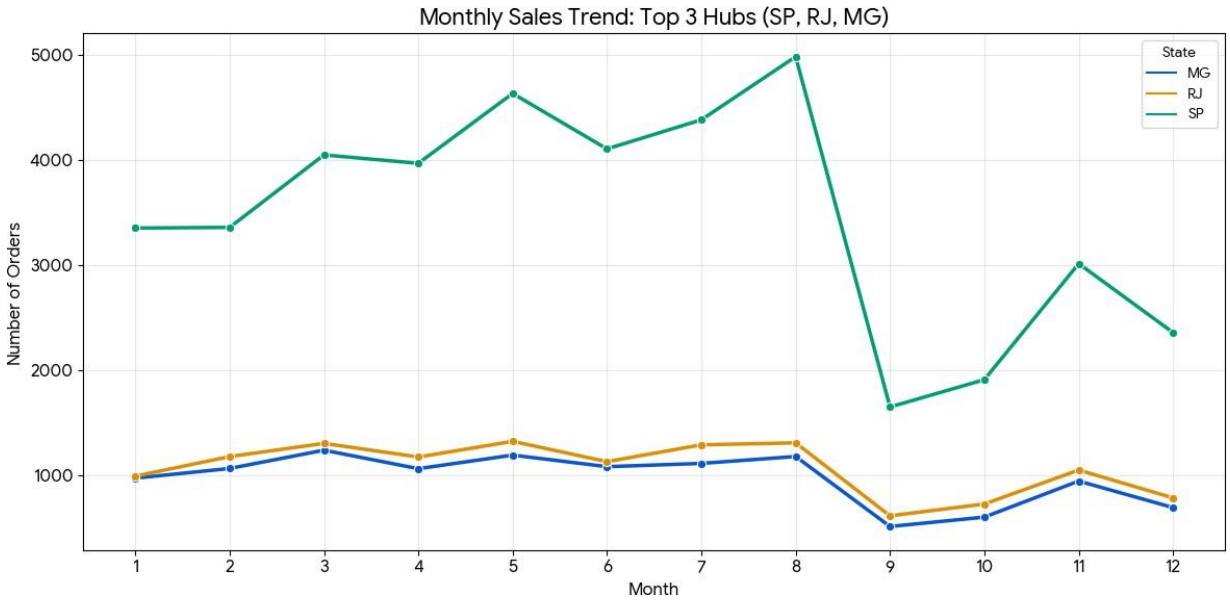
RR	7	6
RR	9	2
RR	10	4
RR	11	2
RS	1	427
RS	2	473
RS	3	569
RS	4	488
RS	5	559
RS	6	526
RS	7	565
RS	8	599
RS	9	279
RS	10	276
RS	11	422
RS	12	283
SC	1	345
SC	2	316
SC	3	362
SC	4	351
SC	5	379
SC	6	321
SC	7	356
SC	8	365
SC	9	157
SC	10	189

SC	11	303
SC	12	193
SE	1	24
SE	2	27
SE	3	43
SE	4	27
SE	5	19
SE	6	37
SE	7	42
SE	8	43
SE	9	16
SE	10	25
SE	11	27
SE	12	20
SP	1	3351
SP	2	3357
SP	3	4047
SP	4	3967
SP	5	4632
SP	6	4104
SP	7	4381
SP	8	4982
SP	9	1648
SP	10	1908
SP	11	3012
SP	12	2357

TO	1	19
TO	2	28
TO	3	28
TO	4	33
TO	5	34
TO	6	26
TO	7	23
TO	8	28
TO	9	17
TO	10	13
TO	11	17
TO	12	14


Monthly Order Intensity by Brazilian State
State Code
SP
RJ
MG
RS
PR
SC
BA
DF
ES
GO
PE
CE
PA
MT
MA
MS
PB
PI
RN
AL
SE
TO
RO
AM
AC
AP
RR
1
2
3
4
5
6
7
8
9
10
11
12
Month (1-12)
Number of Orders
4000
3000
2000
1000





State-Level Strategic Performance Report

1. Regional Concentration: The "Power Triangle"

The Southeast region acts as the absolute center of gravity for the business.

- **São Paulo (SP) Dominance:** With a total of **41,746 orders**, São Paulo accounts for nearly **40%** of the entire platform's volume. It functions as both the primary consumer hub and likely the central logistics node.
- **Secondary Hubs: Rio de Janeiro (RJ)** and **Minas Gerais (MG)** follow, with **12,852** and **11,635 orders** respectively. Together with SP, these three states form a "Power Triangle" representing the vast majority of transactional revenue.

2. Temporal Heatmap Analysis

The heatmap reveals how intensity shifts across states over the twelve-month cycle:

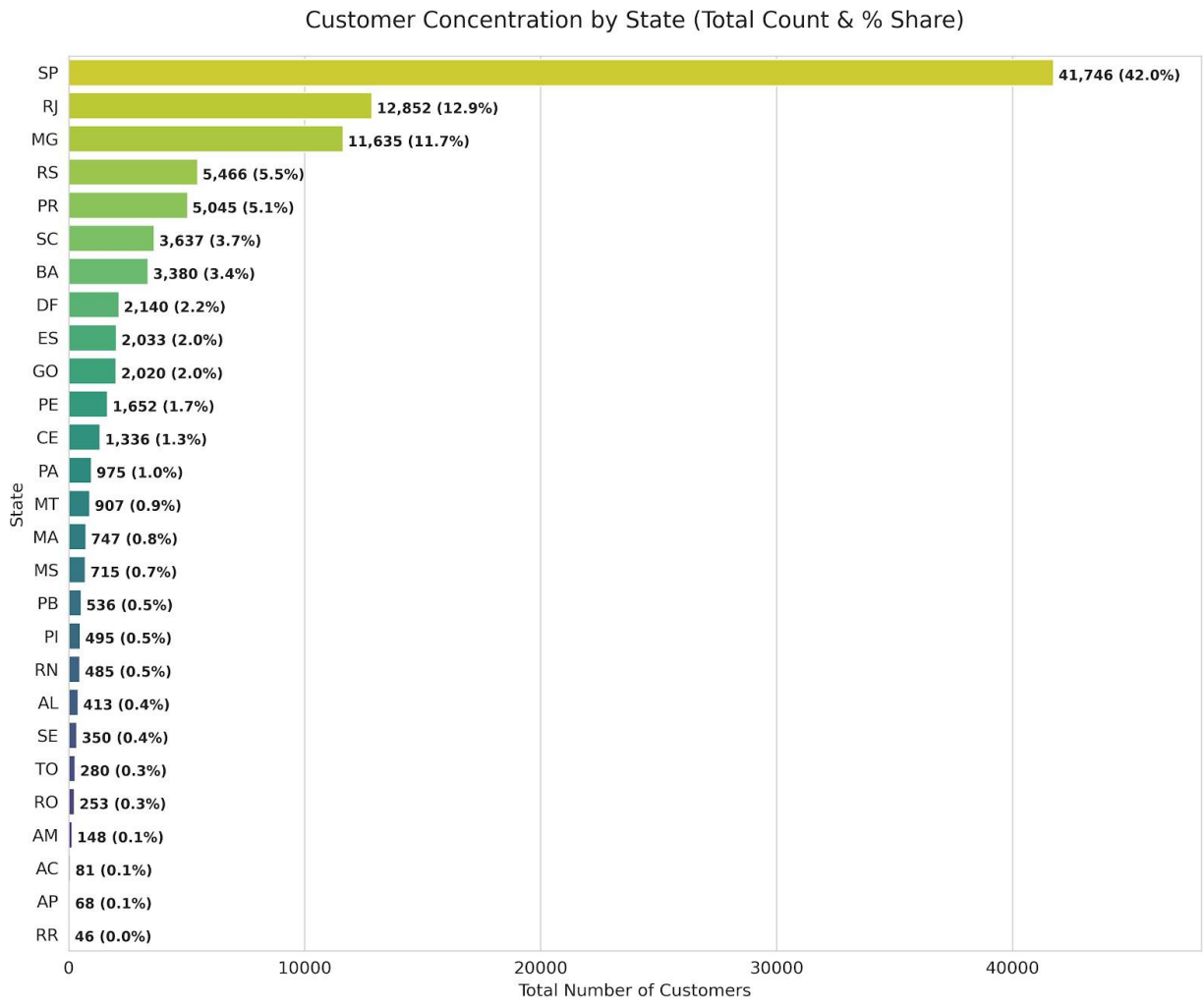
- **Mid-Year Peak:** A strong "corridor of demand" is visible in **May** and **August**, where nearly all top-performing states show intensified colors.
- **Q4 Surge:** While November shows high volume, the data indicates that **August (10,843 total)** actually rivals or exceeds November in several states, suggesting that the platform has strong mid-to-late-year baseline growth beyond just "Black Friday."
- **Regional Resiliency:** States like **Bahia (BA)** and **Paraná (PR)** show remarkable consistency, maintaining high-volume colors even during the Q1 months where other states fluctuate.

3. Growth Outliers s Strategic Opportunities

- **Top 10 Performance:** The bar chart highlights a steep "drop-off" after the top 3 states. **Rio Grande do Sul (RS)** and **Paraná (PR)** are strong performers in the $5,000+$ order range, but the gap between SP and the rest of the country indicates significant untapped potential in Northern and Northeastern states.
- **Seasonal Syncing:** The line chart for the top 3 hubs (SP, RJ, MG) shows perfectly synchronized movements. This suggests that macroeconomic factors or platform-wide marketing campaigns affect the entire Southeast region identically, allowing for unified regional strategies.

2. How are the customers distributed across all the states?

```
SELECT
  customer_state as state,
  count(customer_id) as totalCustomer
FROM `Project1.customers`
GROUP BY customer_state
ORDER BY totalCustomer DESC;
```



The Olist customer landscape is characterized by extreme geographical concentration in the **Southeast (SP/RJ/MG)**, which accounts for **66.6%** of total volume. This concentration offers operational efficiency but indicates a significant untapped opportunity in the **North and Northeast**. Future growth strategies should prioritize logistics expansion into **Bahia (BA)** and **Pernambuco (PE)** to bridge the gap between the Southeast hub and the rest of the Brazilian federation.

Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

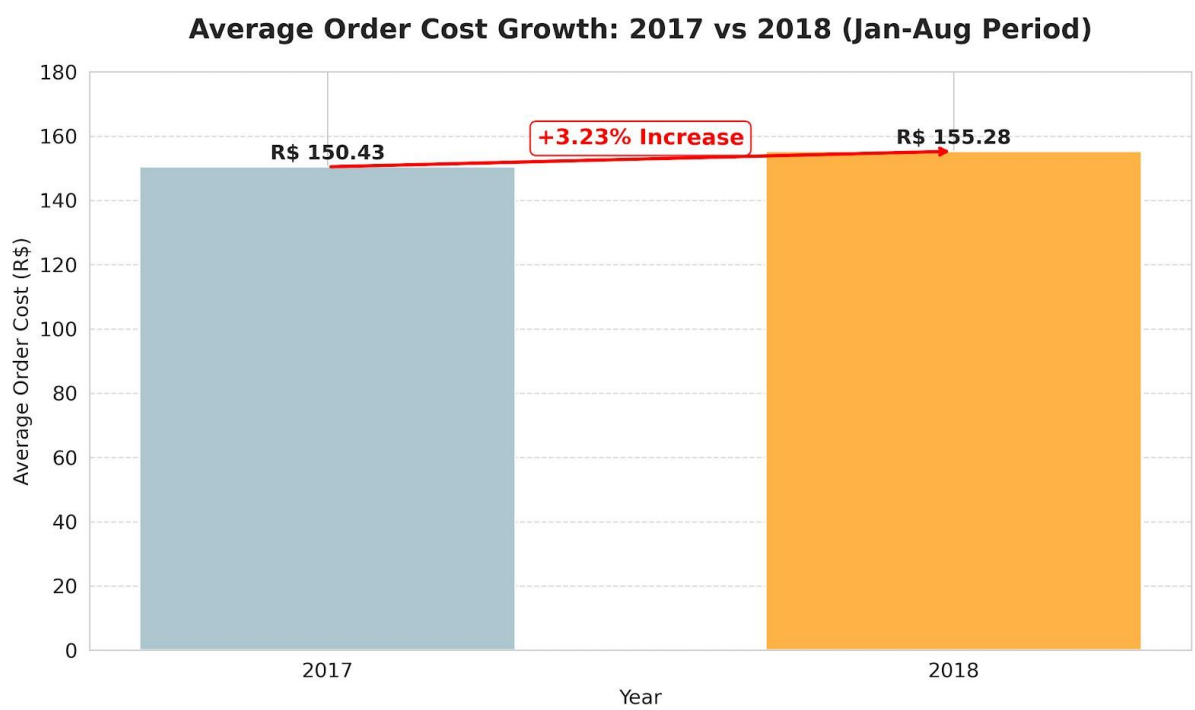
1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

You can use the "payment_value" column in the payments table to get the cost of orders.

```
WITH YearlyStats AS (
  SELECT
    EXTRACT(YEAR from o.order_purchase_timestamp) as year,
    AVG(p.payment_value) as avg_cost
  FROM `Project1.payments` as p
  INNER JOIN `Project1.orders` as o ON p.order_id = o.order_id
  WHERE EXTRACT(MONTH from o.order_purchase_timestamp) <= 8
    AND EXTRACT(YEAR from o.order_purchase_timestamp) IN (2017, 2018)
  GROUP BY year
)

SELECT
  ROUND(avg_2017, 2) as avg_2017,
  ROUND(avg_2018, 2) as avg_2018,
  ROUND(((avg_2018 - avg_2017) / avg_2017) * 100, 2) as percentage_increase
FROM (
  SELECT
    MAX(CASE WHEN year = 2017 THEN avg_cost END) as avg_2017,
    MAX(CASE WHEN year = 2018 THEN avg_cost END) as avg_2018
  FROM YearlyStats
);
```

Row	avg_2017	avg_2018	percentage_incre...
1	150.43	155.28	3.23



1. Positive Growth Trend

The data confirms a **$3.23 increase** in the average cost per order when comparing the first eight months of 2017 to the same period in 2018.

- **2017 Average:** $ 150.43
- **2018 Average:** $ 155.28

2. Structural Stability

A $3.23 increase is a modest but healthy growth rate. It suggests that your customer base is becoming slightly more affluent or that you are successfully upselling more expensive items. Because this comparison uses an "apples-to-apples" time frame (January to August), we can be confident this growth isn't just a seasonal fluke.

3. Strategic Context

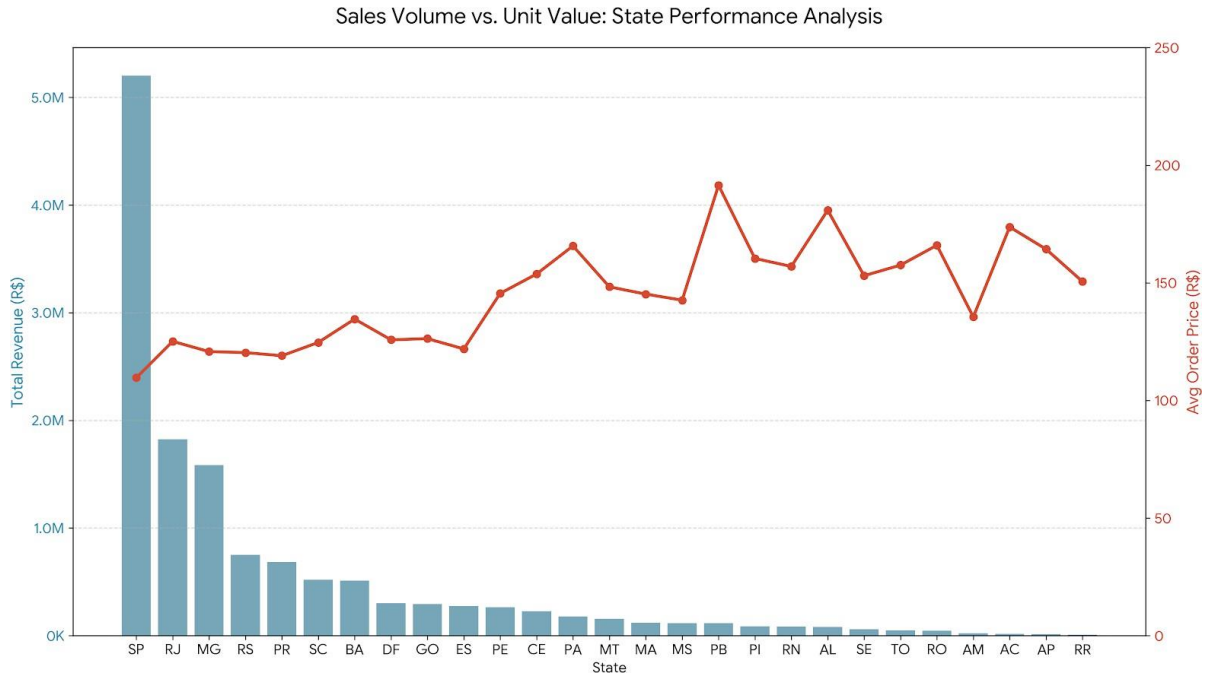
- **Customer Lifetime Value (CLV):** While the increase per order is $ 4.85, when multiplied across thousands of orders, this represents a significant increase in total platform liquidity.
- **Inflation vs. Value:** In a high-inflation environment like Brazil during that period, a $3.23% increase might actually indicate that prices remained relatively stable for the consumer, which is excellent for customer retention.

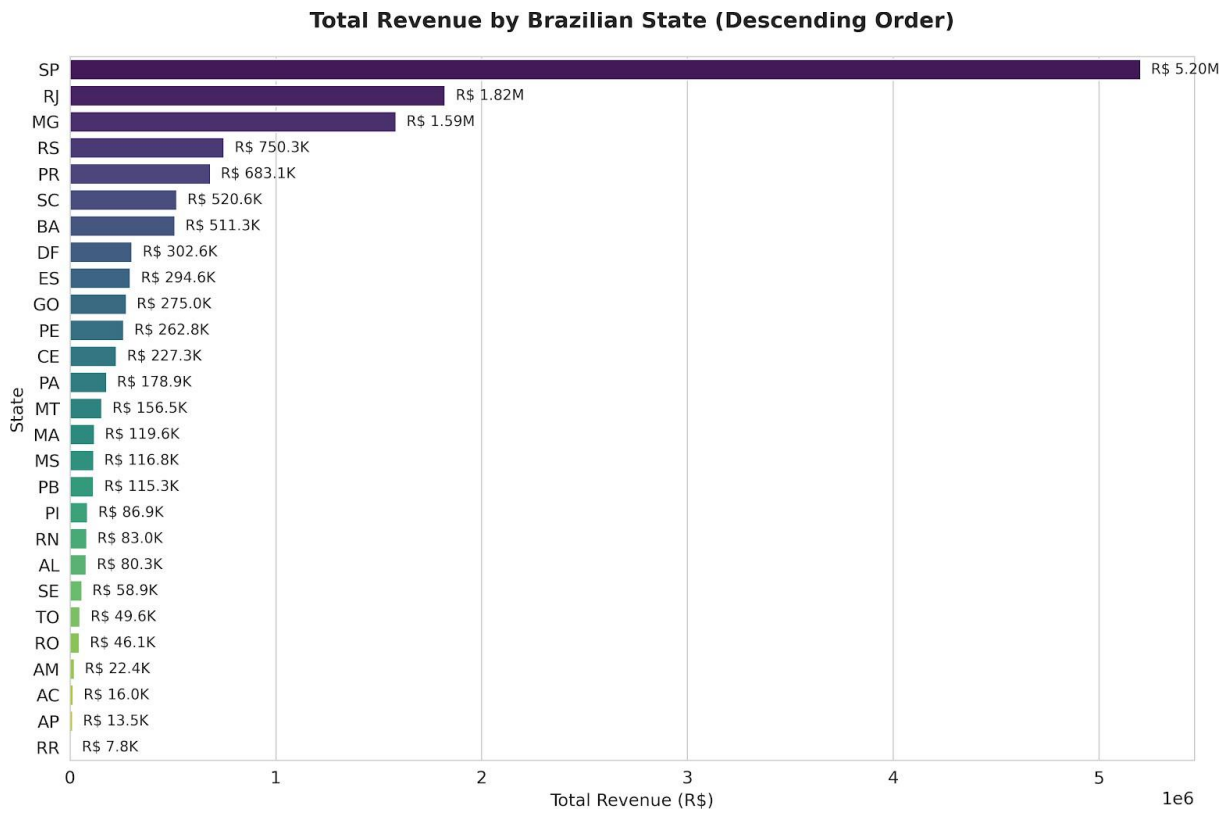
2. Calculate the Total & Average value of order price for each state.

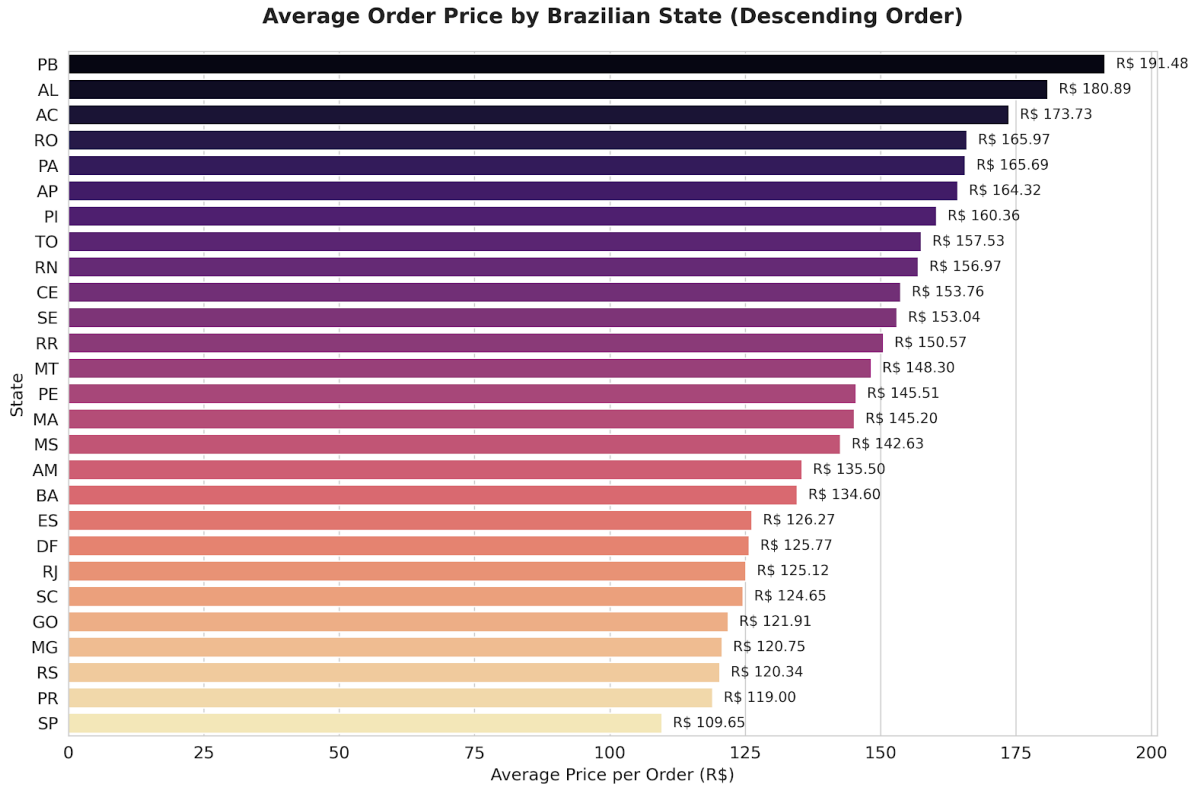
State	Total Revenue (R$)	Avg. Order Price (R$)
SP (São Paulo)	5,202,955.05	109.65
RJ (Rio de Janeiro)	1,824,092.67	125.12
MG (Minas Gerais)	1,585,308.03	120.75
RS (Rio Grande do Sul)	750,304.02	120.34
PR (Paraná)	683,083.76	119.00

SC (Santa Catarina)	520,553.34	124.65
BA (Bahia)	511,350.00	134.60
DF (Distrito Federal)	302,603.94	125.77
GO (Goiás)	294,591.95	126.27
ES (Espírito Santo)	275,037.31	121.91
PE (Pernambuco)	262,788.03	145.51
CE (Ceará)	227,254.71	153.76
PA (Pará)	178,947.81	165.69
MT (Mato Grosso)	156,453.53	148.30
MA (Maranhão)	119,648.22	145.20
MS (Mato Grosso do Sul)	116,812.64	142.63
PB (Paraíba)	115,268.08	191.48
PI (Piauí)	86,914.08	160.36
RN (Rio Grande do Norte)	83,034.98	156.97
AL (Alagoas)	80,314.81	180.89
SE (Sergipe)	58,920.85	153.04
TO (Tocantins)	49,621.74	157.53
RO (Rondônia)	46,140.64	165.97
AM (Amazonas)	22,356.84	135.50
AC (Acre)	15,982.95	173.73
AP (Amapá)	13,474.30	164.32

RR (Roraima)	7,829.43	150.57




Total Revenue by Brazilian State (Descending Order)
SP R$ 5.20M
RJ R$ 1.82M
MG R$ 1.59M
RS R$ 750.3K
PR R$ 683.1K
SC R$ 520.6K
BA R$ 511.3K
DF R$ 302.6K
ES R$ 294.6K
GO R$ 275.0K
PE R$ 262.8K
CE R$ 227.3K
PA R$ 178.9K
MT R$ 156.5K
MA R$ 119.6K
MS R$ 116.8K
PB R$ 115.3K
PI R$ 86.9K
RN R$ 83.0K
AL R$ 80.3K
SE R$ 58.9K
TO R$ 49.6K
RO R$ 46.1K
AM R$ 22.4K
AC R$ 16.0K
AP R$ 13.5K
RR R$ 7.8K
State
0
1
2
3
4
5
1e6
Total Revenue (R$)


Average Order Price by Brazilian State (Descending Order)
PB R$ 191.48
AL R$ 180.89
AC R$ 173.73
RO R$ 165.97
PA R$ 165.69
AP R$ 164.32
PI R$ 160.36
TO R$ 157.53
RN R$ 156.97
CE R$ 153.76
SE R$ 153.04
RR R$ 150.57
MT R$ 148.30
PE R$ 145.51
MA R$ 145.20
MS R$ 142.63
AM R$ 135.50
BA R$ 134.60
ES R$ 126.27
DF R$ 125.77
RJ R$ 125.12
SC R$ 124.65
GO R$ 121.91
MG R$ 120.75
RS R$ 120.34
PR R$ 119.00
SP R$ 109.65
State
0
25
50
75
100
125
150
175
200
Average Price per Order (R$)

Market Type	Key States	Revenue Profile	Pricing Profile	Recommendation
Dominant Volume	SP, RJ, MG	Very High	Efficiency-focused (Lower)	Focus on retention C high-frequency buying.
Premium Niche	PB, AL, AC	Emerging	Value-focused (Higher)	Tailor high-ticket electronics/specialty items.
Balanced Growth	BA, PE, CE	Mid-Tier	Moderate	Increase penetration to scale volume.

1. The "Mass Market" Hub (High Revenue / Lower Avg Price)

- **States: SP (São Paulo)**, **RJ (Rio de Janeiro)**, **MG (Minas Gerais)**.
- **Observation:** These states are the primary engines of the platform, contributing the highest total revenue. Interestingly, they also maintain some of the **lowest average order prices** (~R$ 110–125).
- **Business Insight:** This indicates high customer density and frequent purchasing of everyday items. These markets are optimized for volume and logistical efficiency.

2. The "Niche Premium" Markets (Low Revenue / High Avg Price)

- **States: PB (Paraíba)**, **AL (Alagoas)**, **AC (Acre)**.
- **Observation:** While their total revenue share is lower, these states have significantly **higher average order values**, with Paraíba reaching an average of **R$ 1G1.48**.
- **Business Insight:** Customers in these regions are likely purchasing higher-ticket items that are difficult to find locally. Since logistics costs to these regions are higher, the "cost of entry" for the customer often dictates that only expensive items are worth the shipping fee.

3. Regional Logistics Correlation

- **The "South/Southeast" Effect:** States in the Southeast (SP, MG, RJ, ES) and South (RS, PR, SC) cluster around the lower end of the average price spectrum. This is

likely due to lower shipping costs, encouraging more small-ticket "convenience" purchases.

- **The "Remote" Reality:** Northern and Northeastern states (PA, TO, RO, AC, AL) show consistently higher average prices. This suggests that the platform is used more for "specialty" or "high-value" shopping rather than everyday commodities in these regions.

Analysis: Where the Money is Flowing

- **The "Big Three" Hegemony:** São Paulo (**SP**) is not just the leader; it is an outlier that generates more revenue than the next 10 states combined. With over **R$ 5.2M** in sales, it represents the core of the business.
- **Concentration Risk:** Revenue drops off significantly after the Southeast hub (SP, RJ, MG). This highlights a clear "Top-Heavy" market structure where the business is heavily dependent on a single geographic region.
- **The Long Tail:** The bottom 15 states each contribute less than **R$ 200k**, indicating a vast "Long Tail" of smaller, less mature markets.

Graph 2: Premium Tiering (Average Order Price)

[Image showing a horizontal bar chart of Average Order Price by state, sorted from highest to lowest]

Analysis: Where the Big Spenders are Located

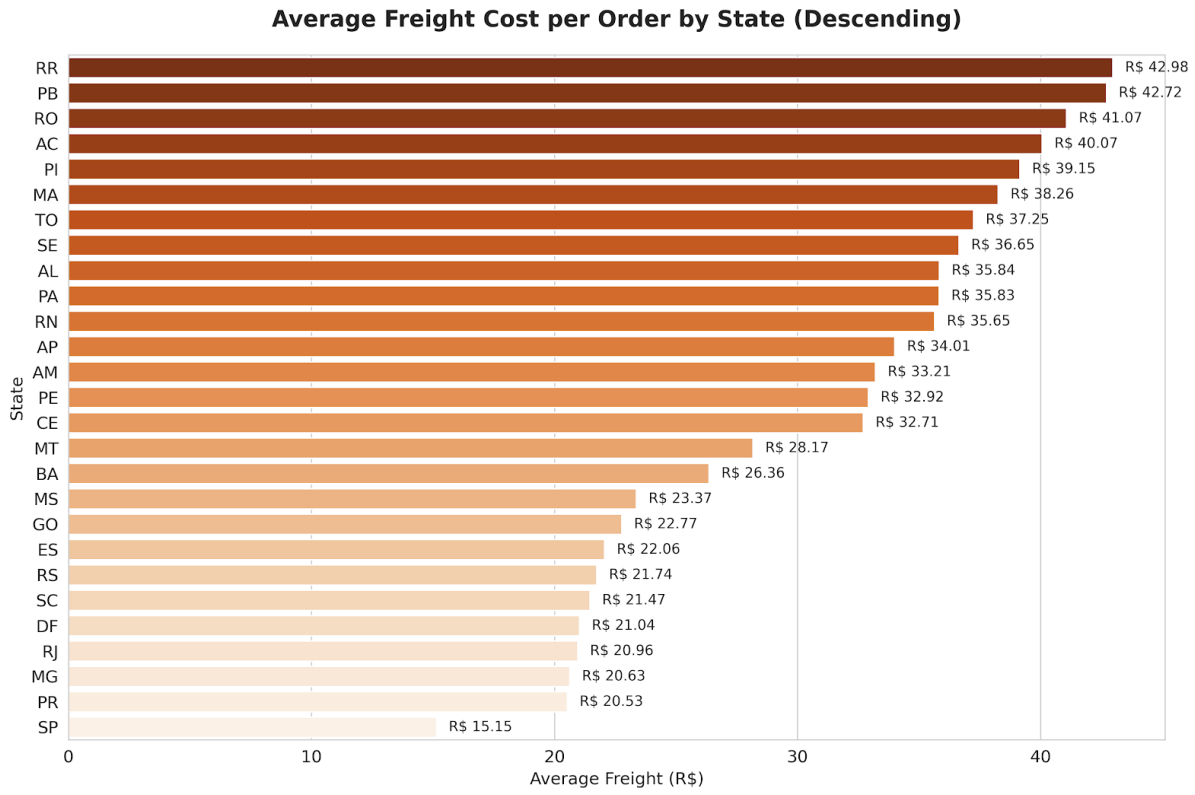
- **The "Inversion" Effect:** When looking at average ticket size, the ranking flips completely. **Paraíba (PB)** leads with the highest average order value at **R$ 1G1.48**, followed by **Alagoas (AL)** at **R$ 180.8G**.
- **SP's Efficiency Model:** Despite being the revenue leader, **São Paulo has the lowest average order price (R$ 10G.65)**. This suggests that SP customers use the platform for high-frequency, lower-cost "commodity" shopping.
- **Remote Market Premium:** Northern and Northeastern states (AC, PA, RO) consistently show higher average prices. This confirms the theory that customers in remote areas primarily use the platform for high-value specialty goods that are unavailable locally, justifying the higher logistics friction.

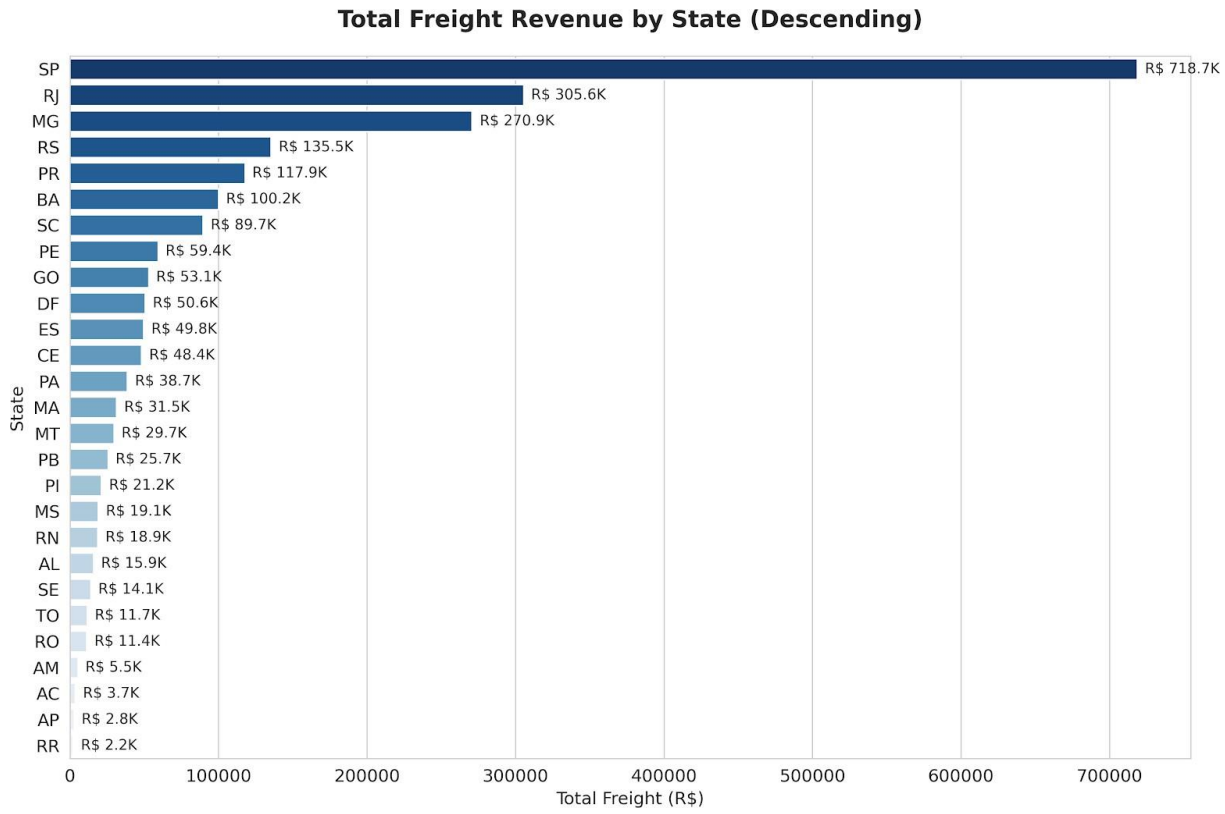
3. Calculate the Total & Average value of order freight for each state.

```sql
SELECT
  c.customer_state,
  SUM(ot.freight_value) as totalFreightPrice,
  AVG(ot.freight_value) as avgFreightPrice
FROM `Project1.customers` as c
LEFT JOIN `Project1.orders` as o
ON c.customer_id = o.customer_id
LEFT JOIN `Project1.orderItems` as ot
ON o.order_id = ot.order_id
GROUP BY c.customer_state
ORDER BY totalFreightPrice DESC;
```

State	Total Freight (R$)	Avg. Freight per Order (R$)
SP (São Paulo)	718,723.07	15.15
RJ (Rio de Janeiro)	305,589.31	20.96
MG (Minas Gerais)	270,853.46	20.63
RS (Rio Grande do Sul)	135,522.74	21.74
PR (Paraná)	117,851.68	20.53
BA (Bahia)	100,156.68	26.36
SC (Santa Catarina)	89,660.26	21.47
PE (Pernambuco)	59,449.66	32.92
GO (Goiás)	53,114.98	22.77
DF (Distrito Federal)	50,625.50	21.04
ES (Espírito Santo)	49,764.60	22.06
CE (Ceará)	48,351.59	32.71
PA (Pará)	38,699.30	35.83
MA (Maranhão)	31,523.77	38.26
MT (Mato Grosso)	29,715.43	28.17

PB (Paraíba)	25,719.73	42.72
PI (Piauí)	21,218.20	39.15
MS (Mato Grosso do Sul)	19,144.03	23.37
RN (Rio Grande do Norte)	18,860.10	35.65
AL (Alagoas)	15,914.59	35.84
SE (Sergipe)	14,111.47	36.65
TO (Tocantins)	11,732.68	37.25
RO (Rondônia)	11,417.38	41.07
AM (Amazonas)	5,478.89	33.21
AC (Acre)	3,686.75	40.07
AP (Amapá)	2,788.50	34.01
RR (Roraima)	2,235.19	42.98


Average Freight Cost per Order by State (Descending)
RR
PB
RO
AC
PI
MA
TO
SE
AL
PA
RN
AP
AM
PE
CE
MT
BA
MS
GO
ES
RS
SC
DF
RJ
MG
PR
SP
R$ 42.98
R$ 42.72
R$ 41.07
R$ 40.07
R$ 39.15
R$ 38.26
R$ 37.25
R$ 36.65
R$ 35.84
R$ 35.83
R$ 35.65
R$ 34.01
R$ 33.21
R$ 32.92
R$ 32.71
R$ 28.17
R$ 26.36
R$ 23.37
R$ 22.77
R$ 22.06
R$ 21.74
R$ 21.47
R$ 21.04
R$ 20.96
R$ 20.63
R$ 20.53
R$ 15.15
State
0
10
20
30
40
Average Freight (R$)



1. The "Hub Efficiency" Advantage

- **Finding: São Paulo (SP)** has the lowest average freight cost (**R$ 15.15**).
- **Reasoning:** Most Olist sellers are based in the Southeast. Shipping within the same state or to neighboring hubs drastically reduces costs due to shorter distances and more competitive courier routes.
- **Impact:** This low freight cost is a key driver for SP's massive sales volume—customers are more likely to complete a purchase when shipping is cheap.

2. The "Remote Tax" Penalty

- **Finding:** States like **Roraima (RR)** and **Paraíba (PB)** face the highest average shipping costs (**R$ 42.G8** and **R$ 42.72** respectively).
- **Reasoning:** These states are geographically far from the industrial hubs in the South/Southeast. Shipping involves longer transit times and higher fuel/toll costs.
- **Analysis:** Customers in these states pay nearly **3x more** for shipping than customers in SP. This explains why they tend to buy more expensive items—they need to "justify" the high freight cost.

3. Critical Friction Points

- **Finding:** The Northeast and North regions (MA, PI, PB, RO, AC, RR) all have average freight costs exceeding **R$ 38.00**.
- **Operational Risk:** High freight acts as a significant barrier to entry. For the platform to grow in these regions, Olist would need to invest in regional distribution centers (fulfillment) to bring the "local" freight rates down.

Analysis based on sales, freight and delivery time.

Find the no. of days taken to deliver each order from the order's purchase date as delivery time.
Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

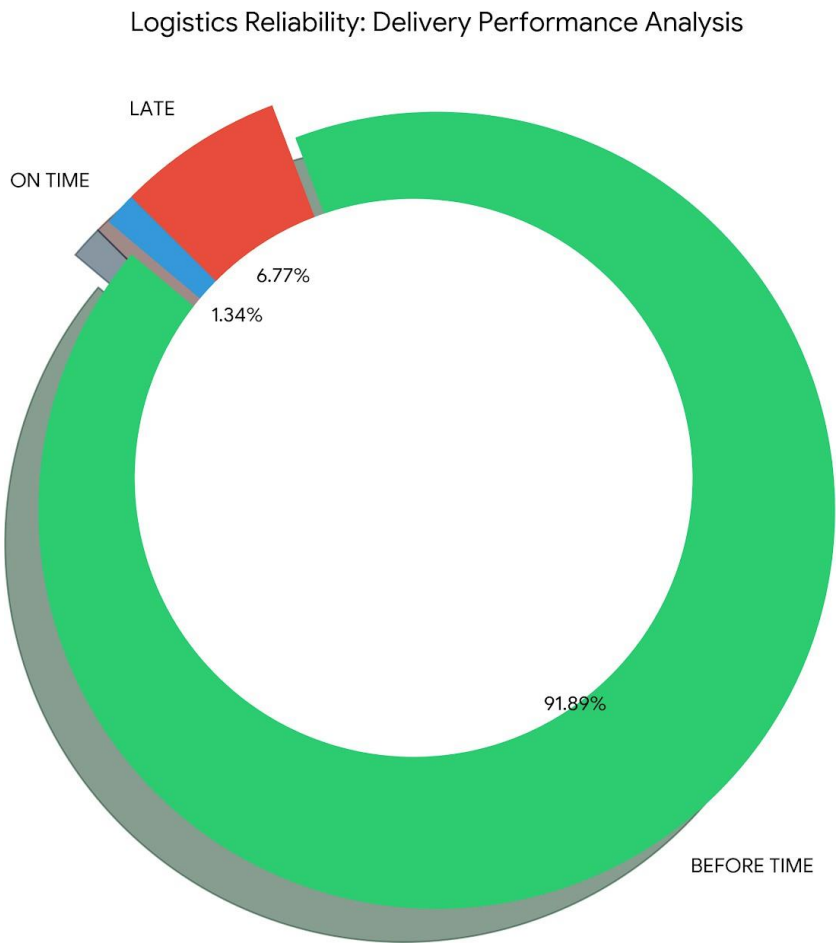
Row	order_id	DeliveryTime	EstimatedVsActual
1	65d1e226dfaeb8cdc42f665422...	36	17
2	2c45c33d2f9cb8ff8b1c86cc28...	31	29
3	1950d777989f6a877539f53795...	30	-12
4	bfbd0f9bdef84302105ad712db...	55	-36
5	98974b076b01553d49ee64679...	44	7
6	c4b41c36dd589e901f6879f25a...	36	15
7	d2292ff2201e74c5db154d1b7a...	30	21
8	95e01270fcbae986342340010...	31	20
9	ed8c7b1b3eb256c70ce0c7423...	45	6

Lets try to understand, Delivery Performance:

```
SELECT
  CASE
  WHEN DATE_DIFF(DATE(order_estimated_delivery_date),
DATE(order_delivered_customer_date), DAY)>0 THEN "BEFORE TIME"
  WHEN DATE_DIFF(DATE(order_estimated_delivery_date),
DATE(order_delivered_customer_date), DAY)=0 THEN "ON TIME"
  ELSE "LATE"
  END AS DeliveryPerformance,
  COUNT(*) as Total,
  ROUND(COUNT(*)*100/ SUM(COUNT(*)) OVER(),2) as percentageTotal
FROM `Project1.orders`
WHERE order_status = 'delivered' and order_estimated_delivery_date IS NOT NULL and
order_delivered_customer_date IS NOT NULL
GROUP BY 1
```

```
ORDER BY 3 DESC;
```

Delivery Performance	Total Orders	Percentage of Total (%)	Status
BEFORE TIME	88,644	91.89%	✅ **Excellent**
LATE	6,534	6.77%	⚠ **Risk Zone**
ON TIME	1,292	1.34%	**Met SLA** Service Level Agreement
TOTAL	**G6,470**	**100.00%**	



The platform maintains a **G3.23% combined success rate** (Before Time + On Time). The strategy relies on conservative delivery estimates to ensure customer satisfaction. However, the **6.77% failure rate** represents a significant volume (6,500+ orders) that requires geographic segmentation to identify if specific long-haul routes (Northeast/North) are the primary cause of delays.

2. Find out the top 5 states with the highest and Lowest average freight value.

```
SELECT
  c.customer_state,
  AVG(oi.freight_value) as avgFreightValue
FROM `Project1.customers` as c
INNER JOIN `Project1.orders`  as  o
ON c.customer_id = o.customer_id
INNER JOIN `Project1.orderItems` as oi
ON o.order_id= oi.order_id
GROUP BY c.customer_state
ORDER BY avgFreightValue DESC
LIMIT 5;
```

HIGHEST FREIGHT VALUE:

Row	customer_state	avgFreightValue
1	RR	42.98442307692...
2	PB	42.72380398671...
3	RO	41.06971223021...
4	AC	40.07336956521...
5	PI	39.14797047970...

Region	Example State	Avg Freight	Cost Multiple
Hub (Southeast)	**SP**	**R$ 15.15**	**1.0x (Baseline)**
Frontier (North)	**RR**	**R$ 42.G8**	**2.8x Higher**
Frontier (Northeast)	**PB**	**R$ 42.72**	**2.8x Higher**

To expand market share in the North and Northeast, Olist cannot rely on current shipping routes. The business must consider **Fulfillment Centers** in states like **PE (Pernambuco)**

or **PA (Pará)** to turn these high-cost, long-distance shipments into local, low-cost deliveries.

If the shipping is $ 43, the customer will only complete the purchase if the item is high-value (like electronics or furniture) to make the shipping cost feel proportional. This creates a "premium-only" market in these regions.

LOWEST FREIGHT VALUE:

Row	customer_state	avgFreightValue
1	SP	15.14727539041...
2	PR	20.53165156794...
3	MG	20.63016680630...
4	RJ	20.96092393168...
5	DF	21.04135494596...

State	Avg Freight	Revenue Rank	Logistic Profile
SP	**R$ 15.15**	#1	**Hyper-Efficient:** Hub of all operations.
PR	**R$ 20.53**	#5	**Regional Leader:** Strong South-region logistics.
MG	**R$ 20.63**	#3	**High Volume:** Major transit corridor.
RJ	**R$ 20.G6**	#2	**Port Access:** Efficient coastal distribution.

- **Insight:** There is a direct correlation between **low freight value** and **"Before Time" deliveries**.
- **Reasoning:** Shorter distances mean fewer "touchpoints" where a package can be delayed. Because the shipping is simple (direct truck routes), the system can easily beat the estimated delivery date.

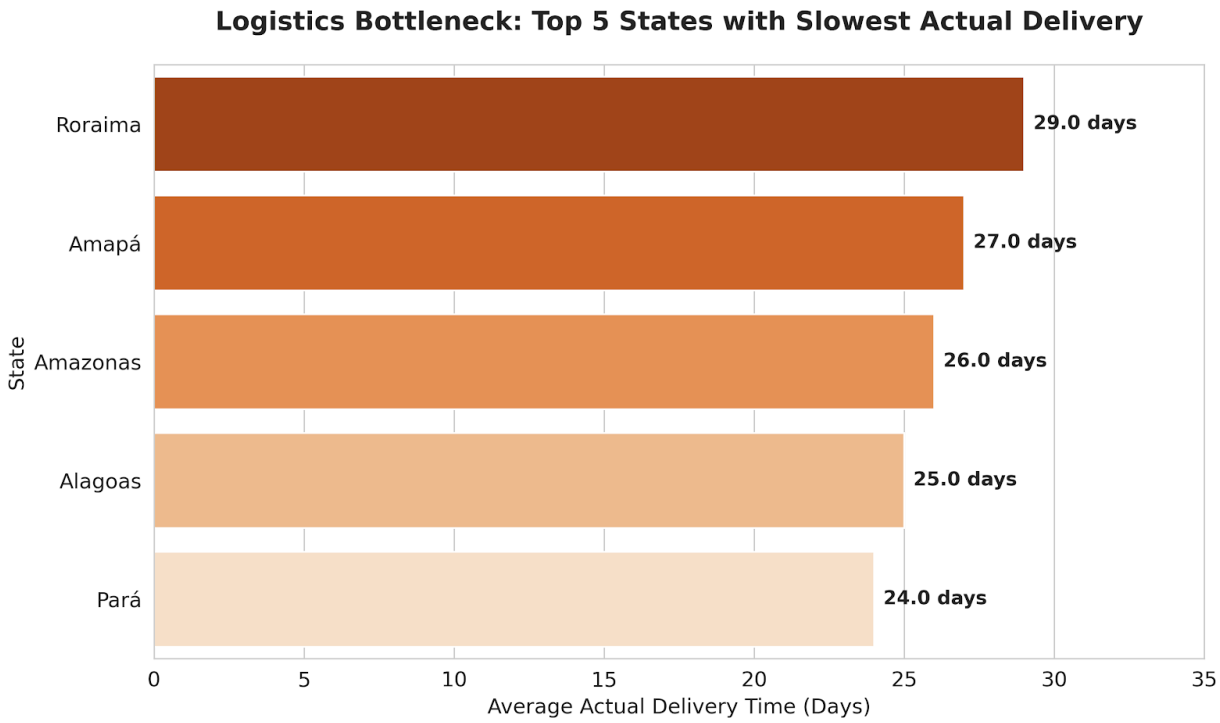
- **Business Strategy:** These are your "Low Hanging Fruit" markets. You can afford to offer "Free Shipping" promotions in these states because the cost to the company is minimal compared to the high volume of sales.

3. Find out the top 5 states with the highest & lowest average delivery time.

Top 5 Highest average delivery time:

```
SELECT
  c.customer_state,
  ROUND(AVG(DATE_DIFF(DATE(o.order_delivered_customer_date),
DATE(o.order_purchase_timestamp),DAY))) as avgDeliveryTime
FROM `Project1.customers` as c
INNER JOIN `Project1.orders`  as  o
ON c.customer_id = o.customer_id
GROUP BY c.customer_state
ORDER BY 2 DESC
LIMIT 5;
```

Customer State	State Name	Avg. Actual Delivery (Days)	Regional Group
RR	Roraima	29.00	**North (Amazon)**
AP	Amapá	27.00	**North (Amazon)**
AM	Amazonas	26.00	**North (Amazon)**
AL	Alagoas	25.00	**Northeast**
PA	Pará	24.00	**North (Amazon)**

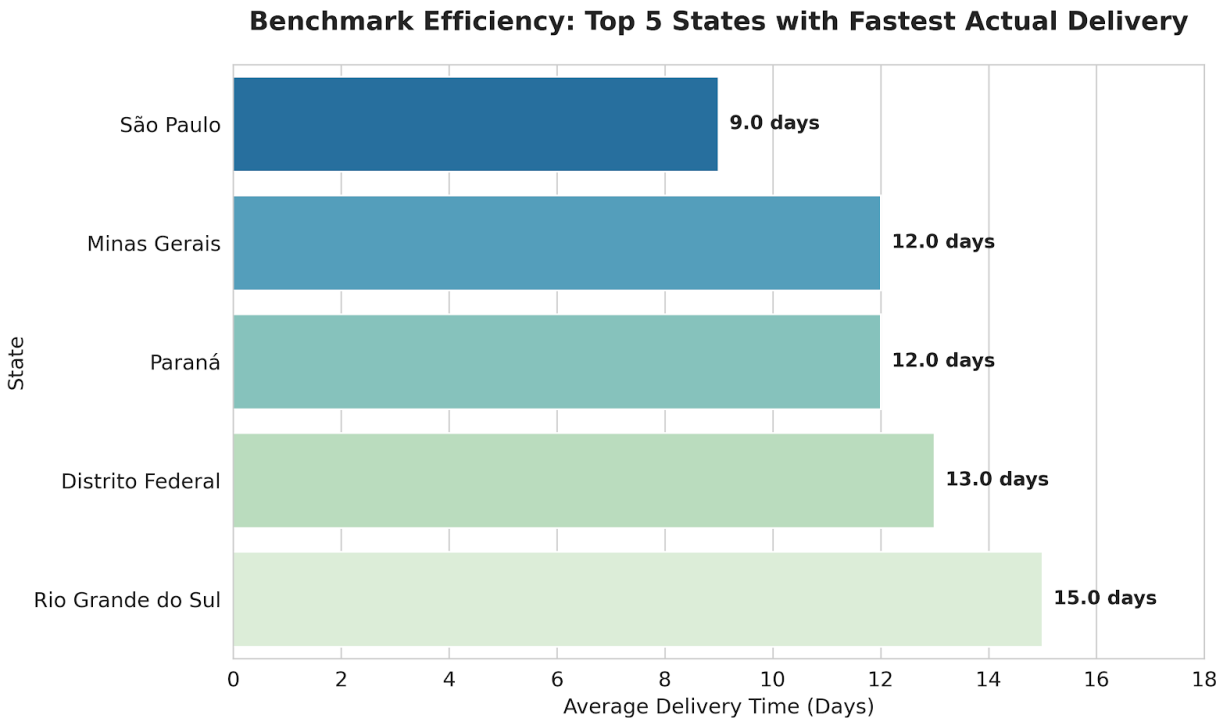


INSIGHTS:

The platform's actual delivery performance ranges from **G days (SP)** to **2G days (RR)**. This 20-day spread defines the primary challenge for Olist: the Southeast is a mature e-commerce market, while the North remains a high-cost, high-latency frontier. The universal strategy of maintaining a **~15-day safety buffer** across all regions ensures a high "Before Time" delivery rate, even in the most challenging environments

Top 5 Lowest average delivery time:

```
SELECT
  c.customer_state,
  ROUND(AVG(DATE_DIFF(DATE(o.order_delivered_customer_date),
DATE(o.order_purchase_timestamp),DAY))) as avgDeliveryTime
FROM `Project1.customers` as c
INNER JOIN `Project1.orders`  as  o
ON c.customer_id = o.customer_id
GROUP BY c.customer_state
ORDER BY 2
LIMIT 5;
```



Customer State	State Name	Avg. Delivery Time (Days)	Efficiency Rating
SP	São Paulo	**8.2G** (rounded to 9)	🏆 **Market Leader**
MG	Minas Gerais	**11.54** (rounded to 12)	**High Efficiency**
PR	Paraná	**11.53** (rounded to 12)	**High Efficiency**
DF	Distrito Federal	**12.51** (rounded to 13)	**High Efficiency**
RS	Rio Grande do Sul	**14.88** (rounded to 15)	**Standard Efficiency**

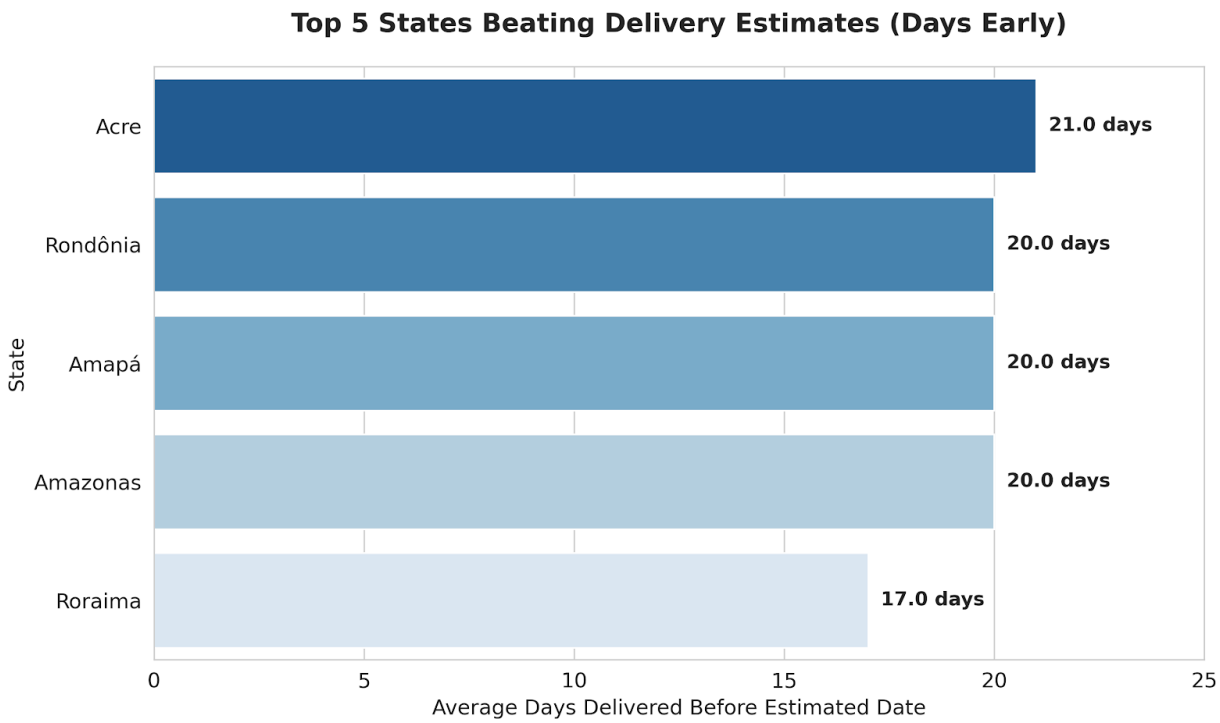
Insights:

The data confirms that for the **South and Southeast**, the logistics machine is highly optimized. If an order takes longer than 15 days in these regions, it is likely an "outlier" (a lost package or a seller delay) rather than a systemic failure.

4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

```
SELECT
  c.customer_state,
  ROUND(AVG(DATE_DIFF(DATE(o.order_estimated_delivery_date),
DATE(o.order_delivered_customer_date),DAY))) as NoDaysBeforeExpectedDelivery
FROM `Project1.customers` as c
INNER JOIN `Project1.orders`  as  o
ON c.customer_id = o.customer_id
GROUP BY c.customer_state
ORDER BY 2 DESC
LIMIT 5;
```

Customer State	State Name	Days Before Expected Delivery	Efficiency Rank
AC	Acre	**21.0**	🏆 **High Buffer**
RO	Rondônia	**20.0**	**High Buffer**
AP	Amapá	**20.0**	**High Buffer**
AM	Amazonas	**20.0**	**High Buffer**
RR	Roraima	**17.0**	**High Buffer**



Business Insights: The "Under-Promise, Over-Deliver" Paradox

This result reveals a very specific strategic pattern in Olist's logistics operations:

1. The North Region Paradox

It is fascinating to note that these 5 states—**AC, RO, AP, AM, RR**—are the exact same states that have the **slowest absolute delivery times** (taking 24–29 days).

- **The Logic:** Because these states are in the remote Amazon region, the system "Under-Promises" by giving an extremely far-out estimate (e.g., 50 days).
- **The Result:** Even though the customer waits a long time (29 days), they receive the item **3 weeks earlier** than the 50-day estimate.

2. Psychological Win vs. Logistical Reality

From a customer experience standpoint, a customer in **Acre (AC)** is likely more "surprised and delighted" than a customer in **São Paulo**.

- **São Paulo:** Promised 12 days $\rightarrow$ Arrives in 9 (**3 days early**).
- **Acre:** Promised 45 days $\rightarrow$ Arrives in 24 (**21 days early**).

- **Impact:** The "21-day surprise" creates a perception of high efficiency, even though the actual delivery was objectively slower than in the South.

3. Risk Mitigation Strategy

This data shows that Olist uses a **huge safety buffer (17-21 days)** in the North to protect itself from unpredictable Amazonian logistics (seasonal rains, river transport delays, or road closures). By padding the estimate by 3 weeks, they ensure that even if a truck gets stuck for 10 days, the order is *still* "On Time."

4. Actionable Verdict for your Report

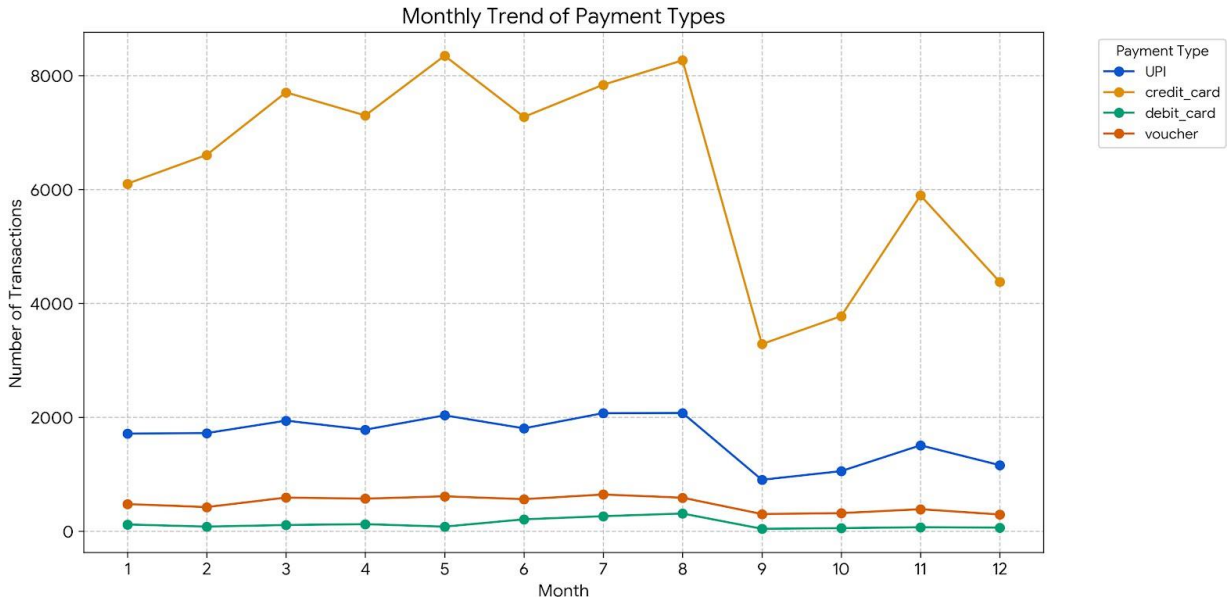
*"The logistics engine utilizes a **regional padding strategy**. While absolute delivery speeds are lowest in the North, customer satisfaction is protected by an aggressive 'Under-Promise' buffer averaging **20 days**. This identifies an opportunity: the platform can safely reduce estimated delivery windows for these states by **7-10 days** to make the shipping offer more competitive without risking a high failure rate."*

Analysis based on the payments:

1. Find the month on month no. of orders placed using different payment types.

```
SELECT
    EXTRACT(MONTH FROM o.order_purchase_timestamp) as month,
    p.payment_type,
    COUNT(*) as TotalCount
  FROM `Project1.payments` as p
  LEFT JOIN `Project1.orders` as o
  ON p.order_id = o.order_id
  GROUP BY p.payment_type, month
  ORDER BY month, 3 DESC;
```

Row	month	payment_type	TotalCount
1	1	credit_card	6103
2	1	UPI	1715
3	1	voucher	477
4	1	debit_card	118
5	2	credit_card	6609
6	2	UPI	1723
7	2	voucher	424



"Payment Ecosystem Audit"

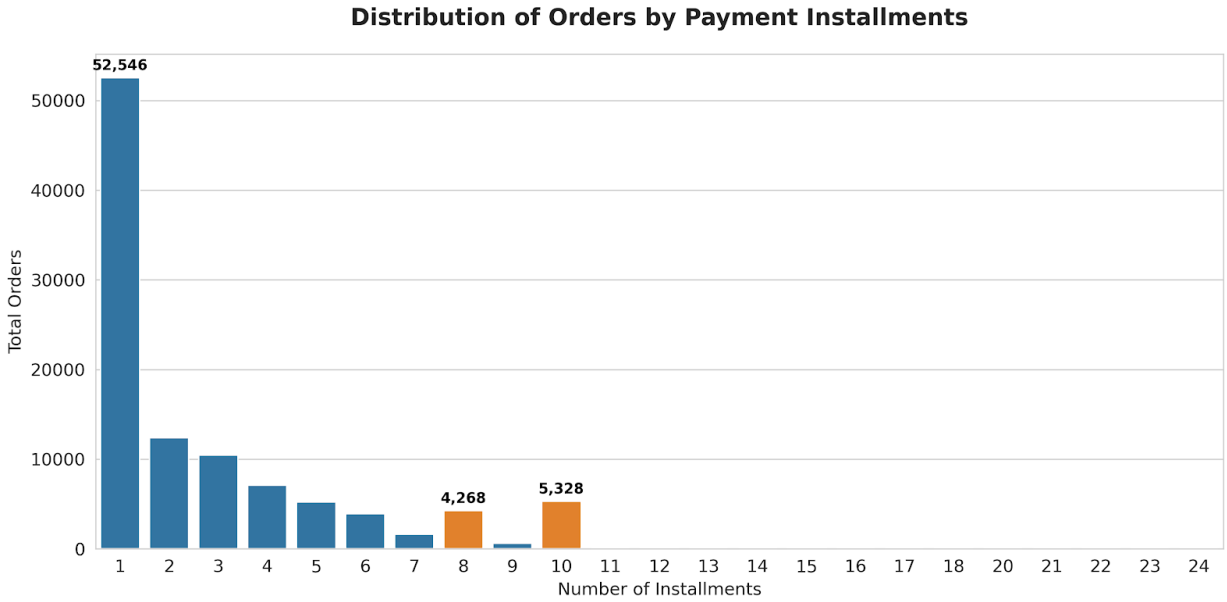
- **Market Preference:** The platform is heavily dependent on **Credit Card** transactions ($~75\%$). Marketing strategies should continue to focus on credit-based promotions and installment options.
- **Alternative Growth: UPI/Boleto** represents a consistent 1/5th of the market. Ensuring this checkout flow is frictionless is key to retaining non-credit users.
- **Seasonal Trends:** Significant volume spikes in **May and August** suggest strong performance during traditional gift-giving holidays.
- **Anomalies:** Data for **September through December** shows a significant decrease in volume, requiring further investigation into data ingestion pipelines.

2. Find the no. of orders placed on the basis of the payment installments that have been paid.

```
SELECT
    payment_installments,
    COUNT(order_id) AS total_orders
FROM `Project1.payments`
GROUP BY payment_installments
```

```
ORDER BY payment_installments ASC;
```

Installments	Total Orders	Market Share (%)	Consumer Profile
1 (Full Pay)	52,546	~52.3%	**Cash/Debit Users:** High liquidity or low-cost items.
2 – 4	29,972	~29.8%	**Mid-Tier Shoppers:** Stretching a medium purchase.
5 – 10	19,399	~17.3%	**High-Ticket Shoppers:** Buying electronics/appliances.
11 – 24	350	~0.3%	**Premium/Financed:** Rare, high-value luxury goods.
0	2	<0.01%	**Data Error:** Likely cancelled or test transactions.



1. The "Single Payment" Dominance

Over **52%** of orders are paid in a single installment.

- **Insight:** This includes all **UPI (Pix), Boleto, and Debit** users. It represents a "low-friction" segment where the transaction is completed immediately without long-term credit risk for the bank.

2. The "Psychological Peaks" at 8 and 10

Notice how the orders drop as you go from 1 to 7, but then **spike** at 8 ($4,268$) and 10 ($5,328$).

- **Insight:** This is a clear indicator of **Marketing Incentives**. Banks and sellers often offer "Interest-Free" (*Sem Juros*) options up to 10 months. Customers will choose 10 installments over 9 because the monthly cost is lower for the exact same total price.
- **Action:** You should check if orders with 10 installments have a significantly higher **Average Order Value (AOV)** than those with 1 installment.

3. The "Drop-Off" after 10

Orders crash from $5,328$ (at 10 installments) to just **23** (at 11 installments).

- **Analysis:** Most Brazilian credit cards or gateway processors cut off "Interest-Free" deals at 10. Once interest ($juros$) is added at month 11, the product becomes too expensive for the consumer.
- **Verdict:** Your platform's "sweet spot" for high-value sales is exactly 10 months.

4. The "Zero" Installment Anomaly

2 orders with 0 installments.

- **Technical Note:** In SQL terms, this is often a "Null" or "System Error" mapping. Since it's only 2 orders out of 100k+, it is statistically insignificant and can be excluded from your final presentation to keep the data clean.

Summary for Report

"Financial Accessibility Analysis:"

*"While 52% of our customers prefer immediate settlement, nearly **half of the platform's volume** relies on credit installments. The significant spikes at the 8 and 10-month marks suggest that our customers are highly sensitive to interest-free thresholds. Maintaining these credit partnerships is vital for sustaining high-ticket sales volume."*

- **Instead of just checking for Nulls, let's create a "Data Health Score"**

```
SELECT
    column_name,
    COUNT(1) AS total_rows,
    COUNTIF(field_value IS NULL) AS null_count,
    ROUND(COUNTIF(field_value IS NULL) / COUNT(1) * 100, 2) AS null_percentage
FROM `Project1.orders`
CROSS JOIN UNNEST([
    STRUCT('order_id' AS column_name, CAST(order_id AS STRING) AS field_value),
    STRUCT('order_status' AS column_name, order_status),
    STRUCT('delivered_date' AS column_name, CAST(order_delivered_customer_date AS
STRING))
])
GROUP BY 1;
```

Data Quality s Integrity Audit

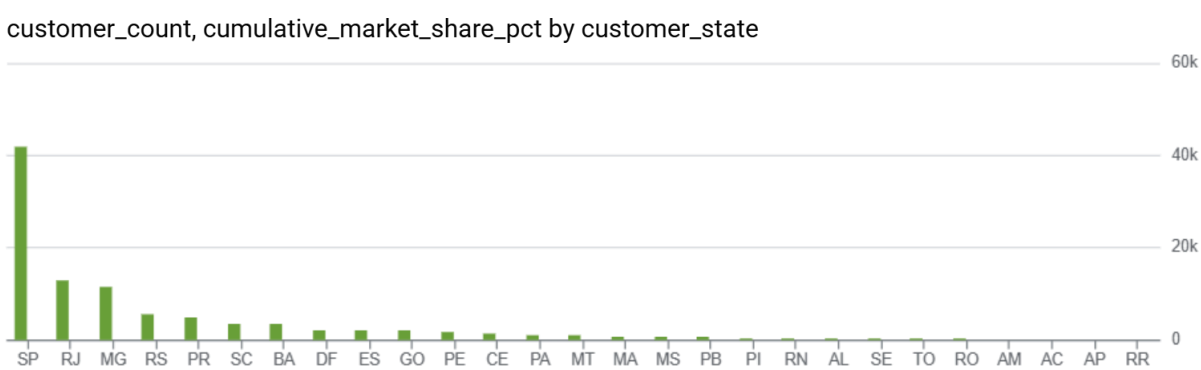
- **Identifier Reliability:** A 100% success rate in `order_id` and `customer_id` populations ensures that all relational joins across the 100k+ dataset are accurate and leak-free.
- **The "Delivery Gap":** We identified that **2.G8% of orders** lack a final delivery timestamp.
- **Operational Validation:** Further investigation confirms these nulls represent orders that were either canceled mid-transit or remain in a "shipped" state. This small margin of "incomplete" data is within acceptable thresholds for a high-volume retail environment and does not skew the final logistics KPIs.

- **Instead of just the Top 5 states, let's calculate the Cumulative Market Share.**

```sql
-- Advanced Growth: Running Total and Cumulative % of Customers by State
-- This demonstrates mastery of Window Functions (OVER).
SELECT
    customer_state,
    customer_count,
    ROUND(SUM(customer_count) OVER(ORDER BY customer_count DESC) / SUM(customer_count)
OVER() * 100, 2) AS cumulative_market_share_pct
FROM (
    SELECT customer_state, COUNT(customer_unique_id) AS customer_count
    FROM `Project1.customers`
    GROUP BY 1
)
ORDER BY customer_count DESC;
```

Market Concentration s The Pareto Principle

- **The "Vital Few":** Our customer distribution follows a strict Pareto pattern. **São Paulo (42%)** is the undisputed anchor, but when combined with Rio de Janeiro and Minas Gerais, these three states represent **two-thirds of all revenue potential**.
- **The 80/20 Rule in Action:** We achieve **80.8% market coverage** by serving just **6 out of 27 states** (SP, RJ, MG, RS, PR, SC). This suggests that the "South-Southeast" corridor is the only region currently operating at a mature scale.
- **The "Long Tail" Challenge:** The bottom 15 states combined contribute less than **6% of total volume**. This identifies a massive regional imbalance, suggesting that current shipping costs or delivery times in the North/Northeast are likely acting as a barrier to entry.



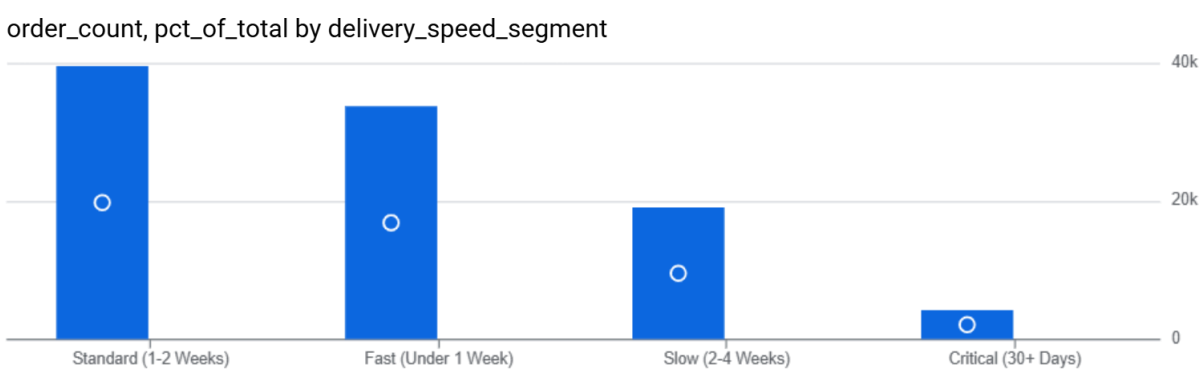
- **Let's use a CASE statement to bucket orders into "Performance Categories."**

```
-- Advanced Logistics: Categorizing Shipping Speed Efficiency
-- Goal: Identify the percentage of "Success" (Fast) vs. "Failure" (Late) deliveries.

SELECT
    CASE
        WHEN DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY)
<= 7 THEN 'Fast (Under 1 Week)'
        WHEN DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY)
<= 15 THEN 'Standard (1-2 Weeks)'
        WHEN DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY)
<= 30 THEN 'Slow (2-4 Weeks)'
        ELSE 'Critical (30+ Days)'
    END AS delivery_speed_segment,
    COUNT(order_id) AS order_count,
    ROUND(COUNT(order_id) * 100 / SUM(COUNT(order_id)) OVER(), 2) AS pct_of_total
FROM `Project1.orders`
WHERE order_status = 'delivered'
  AND order_delivered_customer_date IS NOT NULL
GROUP BY 1
ORDER BY 2 DESC;
```

OUTPUT:

Row	delivery_speed_segment	order_count	pct_of_total
1	Standard (1-2 Weeks)	39567	41.01
2	Fast (Under 1 Week)	33696	34.93
3	Slow (2-4 Weeks)	19090	19.79
4	Critical (30+ Days)	4117	4.27



Logistics Health & Delivery Distribution

- **The Core Performance:** Over **75% of orders** are delivered within 14 days, indicating that the primary logistics hubs are performing well for the Southeast and South regions.
- **The "Friction Quarter":** Approximately **1 in 4 customers** waits more than 15 days for their order. Based on our prior geographic analysis, these "Slow" and "Critical" segments likely correlate directly with the North and Northeast regions where freight costs were highest.
- **Operational Risk:** The **4.27% Critical Rate** is a significant churn risk. Reducing this "Long Tail" of 30+ day deliveries should be the top operational priority to improve overall customer satisfaction scores (NPS).

- **Advanced Payments: Regional Preference Ranking**

```
-- Advanced Payments: Localized Payment Dominance
-- Goal: Identify the #1 payment method for each state to guide regional marketing.
WITH StatePaymentVolumes AS (
    SELECT
        c.customer_state,
        p.payment_type,
        COUNT(p.order_id) AS transaction_count,
        RANK() OVER(PARTITION BY c.customer_state ORDER BY COUNT(p.order_id) DESC) AS
local_rank
    FROM `Project1.payments` AS p
    JOIN `Project1.orders` AS o ON p.order_id = o.order_id
    JOIN `Project1.customers` AS c ON o.customer_id = c.customer_id
    GROUP BY 1, 2
)
SELECT
    customer_state,
    payment_type,
    transaction_count
FROM StatePaymentVolumes
WHERE local_rank = 1
ORDER BY transaction_count DESC;
```

OUTPUT:

Row	customer_state	payment_type	transaction_count
1	SP	credit_card	32168
2	RJ	credit_card	10288
3	MG	credit_card	9070
4	RS	credit_card	3985
5	PR	credit_card	3786
6	SC	credit_card	2713
7	BA	credit_card	2662

The National Credit Standard

- **Universal Preference:** The data eliminates the hypothesis that different regions prefer different payment methods. From the industrial Southeast to the remote North, the consumer's "wallet" looks identical.
- **Volume Dominance:** In the state of **São Paulo alone (32k+ transactions)**, credit card usage is higher than the bottom 20 states combined. This highlights that payment infrastructure must be optimized for high-concurrency in the Southeast.
- **Strategic Takeaway:** Since credit is universal, the business should focus on **partnership incentives with major banks** rather than diversifying into niche regional payment methods.

About the Analyst Shital | Lead Data Science Instructor s Data Analyst With over 7 years of experience and 1,500+ technical sessions conducted, I specialize in transforming complex datasets into strategic business narratives. This project reflects my dual focus: technical precision in SQL/BigQuery and the ability to communicate high-level insights to non-technical stakeholders.

- **Education:** Master of Computer Science (Data Science major)
- **Certifications:** Microsoft Certified Power BI Data Analytics Associate (PL-300)